

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

This document develops a three-generator sigma algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$, rationalized electromagnetic units, and signature $(+, -, -, -)$ are used.

Contents

1	The algebra at a glance	4
2	Hermitian and sigma conjugation	5
3	Paravectors	5
3.1	Core operations	5
3.2	Unit vectors and paravectors	7
3.3	Products and quadratic forms	9
3.4	Quadratic identity and eigenvalues	9
3.5	Real products	10
3.6	Complex products	10
3.7	Projection operators	11
3.8	Functions of paravectors	12
4	Transformations and invariants	18
4.1	Similarity, determinant, and norm	19
4.2	Real paravectors and the Lorentz metric	20
4.3	Rotation action and Rodrigues' formula	20
4.4	Boost action and its sign convention	21
5	Differentiated fields	22
5.1	Paravector differential operators	22
5.2	Components and the product rule	22

6	Space–time kinematics	23
6.1	Intervals, velocity, and momentum	23
6.2	Proper acceleration	24
6.3	World lines, clocks, and light cones	26
6.4	Free-particle action	27
7	Electromagnetism	28
7.1	Maxwell’s equations and wave propagation	28
7.2	Potentials and gauge freedom	29
7.3	The electromagnetic field action	30
7.4	Field invariants	30
7.5	Lorentz force from the action	31
7.6	Acceleration and the mass shell	32
7.7	Energy and momentum flux	32
8	Relativistic wave equations	33
8.1	Amplitudes and their pairing	33
8.2	Energy and momentum as differential operators	34
8.3	Klein–Gordon equation	35
8.4	Dirac equation	37
9	Low-energy limits	41
9.1	Exact reduction and the energy recursion	41
9.2	Leading order: Pauli	42
9.3	The same leading order: Schrödinger	43
9.4	The next free order	43
10	Spin and probability	44
10.1	Spin one-half and rotations	44
10.2	Projectors and probabilities	44
10.3	Spin uncertainty	45
11	Rotations and Lorentz boosts	45
11.1	Covariance of the physical equations	46
11.2	Moving frames, clock rates, and lengths	46
11.3	Boosted fields and energy flux	48
11.4	Varying frames and local bases	48

12 Flat spacetime in generalized coordinates	49
13 Magnetic duality and constant-field motion	50
13.1 The imaginary force and magnetic charge	50
13.2 Exact motion in a constant field	51
14 Extrapolation and speculation	51
14.1 Testing a complex momentum equation	52
14.2 Testing complex spacetime and its null set	53
14.3 Separate masses and small couplings	53
14.4 What would be needed for a dark sector?	54
A Pauli-matrix representation	54
A.1 Faithful representation	55
A.2 Conjugations and product order	55
A.3 Scalar part, determinant, and norm	56
A.4 Matrix amplitudes and column spinors	56
A.5 Dirac columns	57
A.6 Four components and the standard Dirac basis	58
A.7 Quaternion subalgebra	59
References	60
Equation reference	62

1 The algebra at a glance

The generators σ_n , $n = 1, 2, 3$, obey ordered multiplication rules. Their products generally do not commute. The panel gives their defining relations and the general complex element developed below.

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations use

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= A_1 B_1 + A_2 B_2 + A_3 B_3, \\ \mathbf{A} \times \mathbf{B} &= \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \\ \mathbf{A}^2 &:= \mathbf{A} \cdot \mathbf{A}. \end{aligned}$$

Thus the square of a vector always denotes its self dot product. The elementary Cartesian vectors used below are the real vectors

$$\mathbf{e}_1 = \{1, 0, 0\}, \quad \mathbf{e}_2 = \{0, 1, 0\}, \quad \mathbf{e}_3 = \{0, 0, 1\}.$$

Thus $\mathbf{e}_k \cdot \boldsymbol{\sigma} = \sigma_k$; the scalar basis element is 1. Unindexed e denotes only Euler's number.

These eight elements form a complete basis, displayed by type as

$$\begin{array}{cccc} 1 & \sigma_1, \sigma_2, \sigma_3 & \mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3 & \mathbf{i} \\ \text{scalar} & \text{vectors} & \text{bivectors} & \text{pseudoscalar} \end{array}$$

Anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}.$$

Thus its order of multiplication does not matter.

For algebra elements X, Y , the commutator is

$$\begin{aligned} [X, Y] &:= XY - YX \\ [Y, X] &= -[X, Y]. \end{aligned}$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k = 1, 2, 3$, define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m, \end{cases}$$

and the Levi-Civita symbol by

$$\epsilon_{nmk} := \begin{cases} +1 & \text{if } (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{if two or more indices are equal.} \end{cases}$$

For $n, m = 1, 2, 3$, the ordered multiplication rule and commutator are

$$\begin{aligned}\sigma_n \sigma_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \\ [\sigma_n, \sigma_m] &= 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k.\end{aligned}\tag{1.1}$$

2 Hermitian and sigma conjugation

Define *Hermitian conjugation* by (2.1) and *sigma conjugation* by (2.2):

$$(\cdot)^{\mathbf{H}} := \text{reverse the order of every product of sigma generators},\tag{2.1}$$

$$(\cdot)^* := \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order}.\tag{2.2}$$

Thus, for two generators,

$$\begin{aligned}(\sigma_m \sigma_n)^{\mathbf{H}} &= \sigma_n \sigma_m, \\ (\sigma_m \sigma_n)^* &= (-\sigma_m)(-\sigma_n).\end{aligned}$$

Both operations act term by term on sums and leave real coefficients fixed. Their actions on the four basis types are

type representative	scalar 1	vector σ_n	bivector $\mathbf{i} \sigma_n$	pseudoscalar $\mathbf{i}$
$(\cdot)^*$	+	−	+	−
$(\cdot)^{\mathbf{H}}$	+	+	−	−
$(\cdot)^{* \mathbf{H}}$	+	−	−	+

For example, on a bivector the signs follow directly:

$$\begin{aligned}(\mathbf{i} \sigma_n)^* &= \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n, \\ (\mathbf{i} \sigma_n)^{\mathbf{H}} &= \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n.\end{aligned}$$

For arbitrary algebra elements Z, Z' , the definitions give the product laws

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}},\tag{2.3}$$

$$(ZZ')^* = Z^* Z'^*.\tag{2.4}$$

The operations commute with each other: $Z^{* \mathbf{H}} = Z^{\mathbf{H} *}$. Appendix A gives both operations in the Pauli-matrix representation, where $^{\mathbf{H}}$ is the usual matrix Hermitian conjugation.

3 Paravectors

3.1 Core operations

Let $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. The real paravectors $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ form the general complex paravector

$$Z = X + \mathbf{i}Y = (a + \mathbf{i}b) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma}.\tag{3.1}$$

Here $S = a + bi \in \mathbb{C}$ and $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$ are its complex scalar and vector coefficients. Define their extraction functions by

$$\text{sc}(Z) := S, \quad (3.2)$$

$$\text{vec}(Z) := \mathbf{V}. \quad (3.3)$$

This notation applies throughout the algebra unless a real specialization is stated. On the coefficients, both conjugations give $S^* = S^H = a - bi$ and $\mathbf{V}^* = \mathbf{V}^H = \mathbf{A} - i\mathbf{B}$; on the paravector, the two conjugations differ in the vector sign:

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.4)$$

$$Z^H = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.5)$$

Their composition defines the adjugate

$$\text{adj}(Z) := Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.6)$$

The trace is twice the scalar part:

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2\text{sc}(Z) = 2S. \quad (3.7)$$

The determinant is the complex scalar

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2. \quad (3.8)$$

For complex Z' , adjugation reverses product order, giving determinant multiplicativity:

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z), \quad (3.9)$$

$$\det(ZZ') = \text{adj}(Z') (\text{adj}(Z) Z) Z' = \det(Z) \det(Z'). \quad (3.10)$$

Since the determinants are scalars, the determinant of any product is unchanged by reordering its factors. For a complex scalar c ,

$$\begin{aligned} \text{adj}(cZ) &= c \text{adj}(Z), \\ \det(cZ) &= c^2 \det(Z). \end{aligned} \quad (3.11)$$

When $\det(Z) \neq 0$, the inverse is

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \quad (3.12)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, with complex coefficients, division is order-sensitive. An unambiguous quotient exists when Z is invertible and the factors commute:

$$\frac{Z'}{Z} := Z' Z^{-1} = Z^{-1} Z' \quad \text{if } \mathbf{V} \times \mathbf{V}' = 0. \quad (3.13)$$

Differentiating $Z^{-1}Z = 1$ gives the inverse rule

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (3.14)$$

For complex Z, Z', Z'' , scalar part obeys the cyclic laws

$$\text{sc}(ZZ') = \text{sc}(Z'Z), \quad (3.15)$$

$$\text{sc}(ZZ'Z'') = \text{sc}(Z'Z''Z) = \text{sc}(Z''ZZ'). \quad (3.16)$$

Multiplying these identities by 2 gives trace cyclicity. Their similarity invariants are derived in Section 4.

The real and imaginary extractions use Hermitian conjugation:

$$\text{re}(Z) := \frac{1}{2}(Z + Z^H) = X, \quad (3.17)$$

$$\text{im}(Z) := \frac{Z - Z^H}{2i} = Y. \quad (3.18)$$

Adjugation evaluates the scalar and vector extractions explicitly:

$$\text{sc}(Z) = \frac{1}{2}(Z + \text{adj}(Z)) = S, \quad (3.19)$$

$$(\text{vec}(Z)) \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - \text{adj}(Z)) = \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.20)$$

The functions re and im also act on complex coefficients. Composing extractions recovers all four real parts:

$$\begin{aligned} a &= \text{re}(S) = \text{sc}(\text{re}(Z)), \\ b &= \text{im}(S) = \text{sc}(\text{im}(Z)), \\ \mathbf{A} &= \text{re}(\mathbf{V}) = \text{vec}(\text{re}(Z)), \\ \mathbf{B} &= \text{im}(\mathbf{V}) = \text{vec}(\text{im}(Z)). \end{aligned}$$

Appendix A reproduces these intrinsic operations with Pauli matrices.

3.2 Unit vectors and paravectors

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. For $\mathbf{C} \in \mathbb{C}^3$, distinguish the square from the positive norm:

$$\begin{aligned} \mathbf{C}^2 &:= \mathbf{C} \cdot \mathbf{C}, \\ \|\mathbf{C}\|^2 &:= \mathbf{C} \cdot \mathbf{C}^*. \end{aligned}$$

For $S \in \mathbb{C}$, set $\|S\|^2 := SS^*$. The positive paravector norm is

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \quad (3.21)$$

All norms denote nonnegative real square roots. Here $\mathbf{u} \in \mathbb{C}^3$ and Z is a complex paravector. Each name below specifies a condition; a letter alone implies none.

- **Vector of square one.**

$$\mathbf{u}^2 = \mathbf{u} \cdot \mathbf{u} = 1. \quad (3.22)$$

For $\mathbf{C}^2 \neq 0$, take $\mathbf{u} = \mathbf{C}/\sqrt{\mathbf{C}^2}$ with either square root. Its norm need not be one. A nonzero vector with $\mathbf{C}^2 = 0$ cannot be normalized this way.

- **Vector of unit norm.**

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}^* = 1. \quad (3.23)$$

For any $\mathbf{C} \neq 0$, take $\mathbf{u} = \mathbf{C}/\|\mathbf{C}\|$; then $\mathbf{u}^2 = \mathbf{C}^2/\|\mathbf{C}\|^2$ need not be one. For real vectors the two conditions coincide: a *real unit vector* has $\mathbf{u}^2 = \|\mathbf{u}\|^2 = 1$. For complex vectors, specify which holds.

- **Unit paravector.** Z has an inverse on both sides:

$$ZZ^{-1} = Z^{-1}Z = 1. \quad (3.24)$$

- **Unitary paravector.** Its Hermitian conjugate is its inverse:

$$Z^H Z = 1. \quad (3.25)$$

Then $Z^H = Z^{-1}$ and $ZZ^H = 1$.

- **Paravector of determinant one.**

$$\det(Z) = 1. \quad (3.26)$$

It is a unit with $Z^{-1} = \text{adj}(Z)$; it need not be unitary. For real Z , $\text{adj}(Z) = Z^*$, so this condition is also $Z^*Z = 1$. This equivalence need not hold for complex Z .

- **Paravector of unit norm.**

$$\|Z\|^2 = 1. \quad (3.27)$$

For $Z \neq 0$, $Z' = Z/\|Z\|$ has unit norm. It need not have an inverse: $(1 + \mathbf{u} \cdot \boldsymbol{\sigma})/\sqrt{2}$ has norm one and determinant zero for any real unit vector $\mathbf{u}$. Unitarity implies unit norm by (3.39).

- **Self-inverse paravector.** Z is its own inverse:

$$Z^2 = 1. \quad (3.28)$$

Then $Z^{-1} = Z$; it is unitary exactly when $Z^H = Z$. Its parts obey $S^2 + \mathbf{V}^2 = 1$ and $2S\mathbf{V} = 0$: thus $Z = \pm 1$ or $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$ with $\mathbf{u}^2 = 1$.

Two complex vectors illustrate why the normalizations differ:

$\mathbf{C}$	$\mathbf{C}^2$	$\ \mathbf{C}\ ^2$
$\{\sqrt{2}, i, 0\}$	1	3
$\{1, i, 0\}$	0	2

Their vectors of unit norm have squares 1/3 and 0, respectively. The physical roles of the paravector conditions are compared in Section 4.

3.3 Products and quadratic forms

Here $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ are real coefficient vectors. The sigma rule (1.1) gives the dot-cross identity

$$\begin{aligned}
 (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\
 &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} i \sigma_k \\
 &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.
 \end{aligned} \tag{3.29}$$

Adding the product with $\mathbf{A}$ and $\mathbf{B}$ interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}).$$

Consequently, for every nonnegative integer n ,

$$\begin{aligned}
 (\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\
 (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\
 (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}.
 \end{aligned}$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} - i\mathbf{A} \times \boldsymbol{\sigma}.$$

Subtracting the two orders gives

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \tag{3.30}$$

Hence

$$\begin{aligned}
 [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2i\mathbf{A} \times \boldsymbol{\sigma}, \\
 [\mathbf{A} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] &= 2\mathbf{A} \times \boldsymbol{\sigma}, \\
 [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.
 \end{aligned}$$

3.4 Quadratic identity and eigenvalues

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Since $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$ and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$,

$$\begin{aligned}
 (Z - S)^2 &= \mathbf{V}^2, \\
 Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0.
 \end{aligned}$$

Using $\text{tr}(Z) = 2S$ and $\det(Z) = S^2 - \mathbf{V}^2$ gives the trace-determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \tag{3.31}$$

Solving (3.31) for a scalar eigenvalue gives

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}. \tag{3.32}$$

Because $\mathbf{V}$ is generally complex, $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0.$$

3.5 Real products

For real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, (3.29) gives

$$\begin{aligned} XY &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ [X, Y] &= 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ X^2 &= (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.33)$$

Their scalar and vector parts are read directly from these equations. The real quadratic identity is (3.31) with $S = a$. If $\mathbf{A} \times \mathbf{B} = 0$, then $XY = YX$; when $\det(X) \neq 0$, also $X^{-1}Y = YX^{-1}$.

3.6 Complex products

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, where $S, S' \in \mathbb{C}$ and $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$. Complex bilinearity extends (3.29) to the general product

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.34)$$

Thus

$$\begin{aligned} \text{sc}(ZZ') &= SS' + \mathbf{V} \cdot \mathbf{V}', \\ \text{vec}(ZZ') &= S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}'. \end{aligned}$$

Equivalently, write $Z = X + \mathbf{i}Y$ and $Z' = X' + \mathbf{i}Y'$ with all four factors real. Their products need not be real, but distributivity gives

$$ZZ' = (XX' - YY') + \mathbf{i}(XY' + YX').$$

Write $Z = (a + \mathbf{i}b) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}$ with real $a, b, \mathbf{A}, \mathbf{B}$, and use primes for the second paravector's real components. The four real components of ZZ' are

$$\begin{aligned} \text{real scalar: } & aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \\ \text{imaginary scalar: } & ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \\ \text{real vector: } & a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \\ \text{imaginary vector: } & a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \end{aligned}$$

Specializing (3.34) gives the quadratic products:

$$\begin{aligned} Z^2 &= (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2\mathbf{i}(ab + \mathbf{A} \cdot \mathbf{B}) \\ &\quad + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \end{aligned} \quad (3.35)$$

$$ZZ^* = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) - 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.36)$$

$$Z^*Z = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) + 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.37)$$

$$ZZ^H = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.38)$$

$$Z^H Z = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.38)$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2\mathbf{i}(ab - \mathbf{A} \cdot \mathbf{B}).$$

Equations (3.35) and (3.36) show $ZZ^* = Z^*Z$ exactly when $\mathbf{A} \times \mathbf{B} = 0$. Equations (3.37) and (3.38) give the same condition for $ZZ^H = Z^H Z$.

3.6.1 Norm and determinant

For the complex coefficients $S = a + bi$ and $\mathbf{V} = \mathbf{A} + i\mathbf{B}$, the norms defined above give

$$\begin{aligned}\|S\|^2 &= a^2 + b^2, \\ \|\mathbf{V}\|^2 &= \mathbf{A}^2 + \mathbf{B}^2.\end{aligned}$$

For the positive norm (3.21), the quadratic products and cyclicity give the Hermitian form

$$\begin{aligned}\|Z\|^2 &= \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(Z Z^H) \\ &= \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*).\end{aligned}\tag{3.39}$$

For complex coefficients, S^2 and $\mathbf{V}^2$ generally differ from these positive norms. Also $\det(Z)$ may be complex. For a complex scalar $c = re^{i\theta}$, with real $r \geq 0$ and θ , scalar scaling gives

$$\begin{aligned}\det(cZ) &= r^2 e^{2i\theta} \det(Z), \\ \|cZ\|^2 &= \|c\|^2 \|Z\|^2 = r^2 \|Z\|^2.\end{aligned}\tag{3.40}$$

The determinant retains the scalar phase by (3.11); the norm loses it by (3.40).

3.7 Projection operators

For a [real unit vector](#) $\mathbf{u}$, the paravector $\mathbf{u} \cdot \boldsymbol{\sigma}$ is Hermitian and [self-inverse](#). Define its two orthogonal projectors

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).\tag{3.41}$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\begin{aligned}\Pi_{\pm}^2 &= \Pi_{\pm}, \\ \Pi_+ \Pi_- &= 0, \\ \Pi_+ + \Pi_- &= 1, \\ \Pi_{\pm}^H &= \Pi_{\pm}, \\ \Pi_{\pm}^* &= \Pi_{\mp}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} &= \pm \Pi_{\pm}.\end{aligned}$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For a second [real unit vector](#) $\mathbf{u}'$,

$$\begin{aligned}(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. Then

$$\begin{aligned}\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \\ \Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \Pi_{\mp}.\end{aligned}$$

For $\Pi'_{\pm} = (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})/2$, all four products are

$$\begin{aligned}\Pi_{\pm}\Pi'_{\pm} &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \pm \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \Pi_{\pm}\Pi'_{\mp} &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \mp \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

For arbitrary real vector increments $d\mathbf{u}$ and $d\mathbf{u}'$, the same dot-cross decomposition gives the geometric determinant identity

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (3.42)$$

Dropping the nonnegative dot-product square in (3.42) gives

$$\sqrt{\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}))} \geq \frac{1}{2} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|. \quad (3.43)$$

Equation (3.42) is Lagrange's dot-cross identity; (3.43) is a geometric bound, not a quantum uncertainty law.

3.7.1 Ladder operators

For arbitrary real unit vectors $\mathbf{u}, \mathbf{a}$ with $\mathbf{u} \cdot \mathbf{a} = 0$, define complex ladder paravectors and the real projectors of (3.41):

$$\begin{aligned} Z_{\pm} &= \frac{1}{2}(\mathbf{a} \pm i\mathbf{u} \times \mathbf{a}) \cdot \boldsymbol{\sigma} \\ \Pi_{\pm} &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (3.44)$$

Then

$$\begin{aligned} [\mathbf{u} \cdot \boldsymbol{\sigma}, Z_{\pm}] &= \pm 2Z_{\pm}, \\ Z_{\pm}^2 &= 0, \\ Z_{\pm}Z_{\mp} &= \Pi_{\pm}, \\ \mathbf{a} \cdot \boldsymbol{\sigma} &= Z_{+} + Z_{-}, \\ (\mathbf{u} \times \mathbf{a}) \cdot \boldsymbol{\sigma} &= (Z_{+} - Z_{-})/i, \\ \mathbf{u} \cdot \boldsymbol{\sigma} &= \Pi_{+} - \Pi_{-}. \end{aligned}$$

The two signs give the maps between projected sectors:

$$\begin{aligned} Z_{\pm}\Pi_{\pm} &= 0, & Z_{\pm}\Pi_{\mp} &= Z_{\pm}, \\ \Pi_{\pm}Z_{\pm} &= Z_{\pm}, & \Pi_{\mp}Z_{\pm} &= 0. \end{aligned}$$

3.8 Functions of paravectors

3.8.1 Spectral rule

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$, $\mathbf{V} \in \mathbb{C}^3$, and $\mathbf{V}^2 \neq 0$. Define the complex scalars $w, \lambda_{\pm}$, complex vector $\mathbf{u}$, and paravector projectors $\Pi_{\pm}$ by

$$\begin{aligned} w &:= +\sqrt{\mathbf{V}^2}, & \lambda_{\pm} &:= S \pm w, \\ \mathbf{u} &:= \mathbf{V}/w, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The root is the principal scalar root. Thus $\mathbf{u}$ is a **vector of square one**; $\mathbf{u} \cdot \boldsymbol{\sigma}$ is **self-inverse**. Its square gives

$$\begin{aligned} \Pi_{\pm}^2 &= \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + 1) = \Pi_{\pm}, \\ \Pi_{+}\Pi_{-} &= \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0, \\ \Pi_{+} + \Pi_{-} &= 1, \\ Z &= S(\Pi_{+} + \Pi_{-}) + w(\Pi_{+} - \Pi_{-}) = \lambda_{+}\Pi_{+} + \lambda_{-}\Pi_{-}. \end{aligned}$$

These projectors need not be Hermitian. The complementary-projector identities give, for every nonnegative integer n ,

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

For a convergent scalar power series $f(z) = \sum_{n=0}^{\infty} c_n z^n$, with $c_n \in \mathbb{C}$, inserting these powers into $f(Z) = \sum_n c_n Z^n$ gives the spectral rule

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.45)$$

The same rule defines chosen analytic branches at the two eigenvalues. Its proof uses only complementary idempotents and therefore applies to any decomposition with these projector identities.

Adjugation exchanges the eigenvalues, so

$$\begin{aligned} \text{adj}(Z) &= \lambda_- \Pi_+ + \lambda_+ \Pi_-, \\ \det(Z) &= \lambda_+ \lambda_-, \\ Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-. \end{aligned}$$

The inverse requires both eigenvalues to be nonzero. Independently choosing the signs of their principal square roots gives

$$\sqrt{Z} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-. \quad (3.46)$$

There are four distinct roots when both eigenvalues are nonzero, and two when one vanishes. Changing w to $-w$ exchanges the eigenvalue–projector pairs and leaves the function unchanged when its branch choices are exchanged with them.

3.8.2 Real paravectors and direct square roots

For real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, with $a \in \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^3$, and $\mathbf{A} \neq 0$, the spectral quantities above reduce to

$$w = \|\mathbf{A}\|, \quad \mathbf{u} = \mathbf{A}/\|\mathbf{A}\|, \quad \lambda_{\pm} = a \pm \|\mathbf{A}\|.$$

The projectors are now Hermitian. In particular, for a **real unit vector** $\mathbf{u}$,

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1) \Pi_+ + f(-1) \Pi_- \\ &= \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1)) \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm \Pi_+ \pm i \Pi_-. \end{aligned}$$

The two root signs are independent. For a fixed analytic branch, $f(X) \rightarrow f(a)$ as $\mathbf{A} \rightarrow 0$; independent root choices require the scalar case below.

A direct component derivation also determines all *real* roots. Let $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ be real, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and seek real $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ such that $X^2 = Y$. Equating parts gives

$$\begin{aligned} a^2 + \mathbf{A}^2 &= b, \\ 2a\mathbf{A} &= \mathbf{B}. \end{aligned}$$

If $\mathbf{B} \neq 0$, then $a \neq 0$ and $\mathbf{A} = \mathbf{B}/(2a)$. Substitution gives

$$\begin{aligned} a^4 - ba^2 + \frac{1}{4}\mathbf{B}^2 &= 0, \\ a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right). \end{aligned} \quad (3.47)$$

For each nonnegative a^2 in (3.47), take both scalar roots a and set $\mathbf{A} = \mathbf{B}/(2a)$; then $\mathbf{A}^2 = b - a^2$. There are four real roots when $b > \|\mathbf{B}\|$, two when $b = \|\mathbf{B}\| > 0$, and none when $b < \|\mathbf{B}\|$.

If $\mathbf{B} = 0$, the equations instead give the scalar roots $X = \pm\sqrt{b}$ and the pure-vector roots $X = \mathbf{A} \cdot \boldsymbol{\sigma}$ with $\mathbf{A}^2 = b$ for $b > 0$. For $b = 0$ only $X = 0$ is real; for $b < 0$ there is no real root.

3.8.3 Coalescing eigenvalues

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 = 0$, define $N = \mathbf{V} \cdot \boldsymbol{\sigma}$, so $N^2 = 0$. A scalar function analytic at S has the terminating Taylor expansion

$$f(Z) = f(S) + f'(S)N. \quad (3.48)$$

For $S \neq 0$, the chosen square root and logarithm are

$$\sqrt{Z} = r + \frac{N}{2r}, \quad (3.49)$$

$$\log Z = \log S + \frac{N}{S}. \quad (3.50)$$

Here $r \in \mathbb{C}$ is a chosen scalar root of S ($r^2 = S$); use a consistent logarithm branch. A nonzero N with $S = 0$ has no square root: such a root would have only zero eigenvalues, and its quadratic identity would force its square to vanish. It has no logarithm because it is not invertible. If $N = 0$, a fixed analytic branch gives $f(S)$; non-scalar square roots of a scalar must instead be found from the component equations $a^2 + \mathbf{A}^2 = S$, $2a\mathbf{A} = 0$, now allowing a and $\mathbf{A}$ to be complex. Independent branch choices need not remain finite as distinct eigenvalues coalesce.

3.8.4 Exponential: projection and series derivations

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 \neq 0$, recall $w = +\sqrt{\mathbf{V}^2}$, the **vector of square one** $\mathbf{u} = \mathbf{V}/w$, and $\lambda_{\pm} = S \pm w$. The spectral rule (3.45) gives

$$\begin{aligned} \exp(Z) &= \frac{1}{2}e^{S+w}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{1}{2}e^{S-w}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^S \left(\frac{e^w + e^{-w}}{2} + \frac{e^w - e^{-w}}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Independently, S commutes with the vector part and $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = w^{2n}$, $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = w^{2n+1}\mathbf{u} \cdot \boldsymbol{\sigma}$. Separating the defining series therefore gives

$$\begin{aligned} \exp(Z) &= e^S \sum_{n=0}^{\infty} \frac{(w\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= e^S \left(\sum_{n=0}^{\infty} \frac{w^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Both derivations give

$$\exp(Z) = e^S (\cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.51)$$

This is independent of the choice w or $-w$: $\cosh$ is even, while both $\sinh w$ and $\mathbf{u}$ change sign. When $\mathbf{V}^2 = 0$, (3.48) instead gives

$$\exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = e^S (1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

The defining series also gives

$$\begin{aligned}\exp(Z^*) &= \exp(Z)^*, \\ \exp(Z^H) &= \exp(Z)^H.\end{aligned}$$

For the real components $a, b, \mathbf{A}, \mathbf{B}$ of $Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, write

$$w = +\sqrt{\mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}}.$$

Substituting $\mathbf{u} = (\mathbf{A} + i\mathbf{B})/w$ into (3.51) gives the component form

$$\exp(Z) = e^a(\cos b + i \sin b) \left(\cosh w + \frac{\sinh w}{w} (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \right). \quad (3.52)$$

Its real and imaginary specializations are

$$\exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) = e^a \left(\cosh \|\mathbf{A}\| + \frac{\sinh \|\mathbf{A}\|}{\|\mathbf{A}\|} \mathbf{A} \cdot \boldsymbol{\sigma} \right), \quad (3.53)$$

$$\exp(i(b + \mathbf{B} \cdot \boldsymbol{\sigma})) = (\cos b + i \sin b) \left(\cos \|\mathbf{B}\| + i \frac{\sin \|\mathbf{B}\|}{\|\mathbf{B}\|} \mathbf{B} \cdot \boldsymbol{\sigma} \right). \quad (3.54)$$

Setting $a = 0$ in (3.53) or $b = 0$ in (3.54) gives the pure-vector exponentials. Every quotient uses its analytic limit at a zero argument.

3.8.5 Rotation and boost factors

Let $\theta \in \mathbb{R}$ and let $\mathbf{u} \in \mathbb{R}^3$ be a **real unit vector**, $\mathbf{u}^2 = 1$. The two special exponentials just derived define the complex rotation factor R and the real boost factor L :

$$R(\theta, \mathbf{u}) := \exp\left(i \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.55)$$

$$L(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh \frac{\theta}{2} + \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.56)$$

In (3.55) and (3.56), the parameter is a rotation angle for R and a rapidity for L ; both are real dimensionless numbers. Here these names specify algebraic objects. Their actions on a general paravector are derived in Section 4; their physical applications follow in Section 11.

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, reversing the parameter gives the inverses

$$R^{-1} = R(-\theta, \mathbf{u}) = \cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.57)$$

$$L^{-1} = L(-\theta, \mathbf{u}) = \cosh \frac{\theta}{2} - \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.58)$$

Sigma conjugation changes both i and $\boldsymbol{\sigma}$ in sign; Hermitian conjugation changes i but fixes a real vector. Hence

$$R^* = R, \quad (3.59)$$

$$R^H = R^{-1}, \quad (3.60)$$

$$L^* = L^{-1}, \quad (3.61)$$

$$L^H = L. \quad (3.62)$$

Equation (3.60) makes R unitary; (3.62) makes L Hermitian, so $L^H L = L^2$. Their determinants and positive norms follow from the circular and hyperbolic identities:

$$\det(R) = \cos^2(\theta/2) + \sin^2(\theta/2) = 1, \quad (3.63)$$

$$\det(L) = \cosh^2(\theta/2) - \sinh^2(\theta/2) = 1, \quad (3.64)$$

$$\|R\|^2 = 1, \quad (3.65)$$

$$\|L\|^2 = \cosh \theta. \quad (3.66)$$

Equations (3.63) and (3.64) make both **paravectors of determinant one**, so their adjugates equal their inverses. Equations (3.65) and (3.66) distinguish their positive norms; only R is unitary for nonzero θ . The half-angle gives the sign and period:

$$R(\theta + 2\pi, \mathbf{u}) = -R(\theta, \mathbf{u}), \quad (3.67)$$

$$R(\theta + 4\pi, \mathbf{u}) = R(\theta, \mathbf{u}). \quad (3.68)$$

Gamma form. Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.56). Define its real velocity parameter and gamma factor by

$$\beta := \tanh \theta, \quad (3.69)$$

$$\gamma := \cosh \theta = \frac{1}{\sqrt{1 - \beta^2}}. \quad (3.70)$$

Here $-1 < \beta < 1$, $\gamma \geq 1$, and $\gamma\beta = \sinh \theta$. The half-angle identities give

$$\begin{aligned} \cosh \frac{\theta}{2} &= \sqrt{\frac{\gamma + 1}{2}}, \\ \sinh \frac{\theta}{2} &= \frac{\gamma\beta}{\sqrt{2(\gamma + 1)}}. \end{aligned}$$

Hence the boost factor itself is

$$L = \sqrt{\frac{\gamma + 1}{2}} \left(1 + \frac{\gamma\beta}{\gamma + 1} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.71)$$

Squaring doubles the exponent, so the full-rapidity paravector is

$$L^2 = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.72)$$

Equation (3.72) has prefactor γ ; L has the half-angle prefactor in (3.71). Equivalently, for any real vector $\mathbf{A} \neq 0$, put $\mathbf{v} = \tanh \|\mathbf{A}\| \mathbf{A}/\|\mathbf{A}\|$ and $\gamma = \cosh \|\mathbf{A}\|$. Then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).$$

The zero-vector case follows by continuity. This last real paravector has **determinant one** and becomes proper velocity in Section 6.

3.8.6 Cayley transform

Let $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ be real, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and $0 < \|\mathbf{B}\| < b$. Set the [real unit vector](#) $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$ and recall $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Its sigma conjugate $Y^* = b - \mathbf{B} \cdot \boldsymbol{\sigma} = \text{adj}(Y)$ commutes with Y , so

$$\frac{Y}{Y^*} = \frac{Y^2}{\det(Y)} = \frac{b^2 + \mathbf{B}^2 + 2b\mathbf{B} \cdot \boldsymbol{\sigma}}{b^2 - \mathbf{B}^2}.$$

Define the real parameter $\theta = 2 \tanh^{-1}(\|\mathbf{B}\|/b)$. The double-angle identities give

$$\begin{aligned} \cosh \theta &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \sinh \theta &= \frac{2b\|\mathbf{B}\|}{b^2 - \mathbf{B}^2}. \end{aligned}$$

Thus the quotient is the full-rapidity exponential

$$\begin{aligned} \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ + e^{-\theta} \Pi_-. \end{aligned} \tag{3.73}$$

Equation (3.73) is $L(\theta, \mathbf{u})^2$ from (3.72). Its real generator vector is $\mathbf{A} = \theta \mathbf{u}$, while the Cayley vector obeys

$$\mathbf{B} = b \tanh\left(\frac{\|\mathbf{A}\|}{2}\right) \frac{\mathbf{A}}{\|\mathbf{A}\|}. \tag{3.74}$$

Thus $\|\mathbf{B}\|/b = \tanh(\theta/2)$, the half-rapidity parameter. The corresponding real gamma factor and vector are

$$\begin{aligned} \gamma &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \mathbf{v} &= \frac{2b\mathbf{B}}{b^2 - \mathbf{B}^2}. \end{aligned}$$

They give the equivalent forms

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{Y}{Y^*}.$$

Choosing $b = 1$ gives the Cayley form with scalar part one; $\mathbf{B} = 0$ follows by continuity.

3.8.7 Composing the factors

Recall $L(\theta, \mathbf{u}) = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $R(\theta, \mathbf{u}) = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.56) and (3.55). For a fixed [real unit direction](#) $\mathbf{u}$ and real parameters θ, θ' , the generators commute, so their parameters add:

$$L(\theta, \mathbf{u})L(\theta', \mathbf{u}) = L(\theta + \theta', \mathbf{u}). \tag{3.75}$$

$$\begin{aligned} R(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta + \theta', \mathbf{u}), \\ L(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta', \mathbf{u})L(\theta, \mathbf{u}). \end{aligned}$$

For the aligned real and imaginary generator, put $w = (\theta + i\theta')/2 \in \mathbb{C}$. Its exponential is

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = L(\theta, \mathbf{u})R(\theta', \mathbf{u}) = \cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Expanding the complex scalar coefficients gives

$$\begin{aligned}\cosh w &= \cosh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \sinh \frac{\theta}{2} \sin \frac{\theta'}{2}, \\ \sinh w &= \sinh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \cosh \frac{\theta}{2} \sin \frac{\theta'}{2}.\end{aligned}$$

This aligned factorization requires parallel real and imaginary generator vectors. For different [real unit directions](#) $\mathbf{u}, \mathbf{v}$, the vector product instead gives

$$\left[\frac{1}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, \frac{1}{2}\mathbf{v} \cdot \boldsymbol{\sigma}\right] = \frac{1}{2}(\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (3.76)$$

This commutator is a rotation generator: boosts about different axes do not commute.

3.8.8 Logarithm

For complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ with $\mathbf{V}^2 \neq 0$, recall $w = +\sqrt{\mathbf{V}^2}$, the [vector of square one](#) $\mathbf{u} = \mathbf{V}/w$, $\lambda_{\pm} = S \pm w$, and $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Assume both eigenvalues are nonzero. For a nonzero complex scalar λ and any integer n , the scalar branches are

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi).$$

Choose one branch for each eigenvalue. The spectral rule gives

$$\begin{aligned}\log Z &= \log(\lambda_+)\Pi_+ + \log(\lambda_-)\Pi_- \\ &= \frac{1}{2}(\log \lambda_+ + \log \lambda_-) + \frac{1}{2}(\log \lambda_+ - \log \lambda_-)\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (3.77)$$

Exchanging w and $-w$ exchanges both eigenvalue–projector pairs and preserves the same chosen logarithm. Independent scalar branches give the different logarithms.

For a [real unit vector](#) $\mathbf{u}$, (3.77) uses eigenvalues 1 and -1 , giving, for integers n, m ,

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \Pi_+ + (2m+1)\pi i \Pi_-. \quad (3.78)$$

For real $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$, $\mathbf{B} \in \mathbb{R}^3$, and $b > \|\mathbf{B}\|$, both eigenvalues are positive. Combining their real logarithms yields

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1}\left(\frac{\|\mathbf{B}\|}{b}\right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.79)$$

The vector quotient uses its limit at $\mathbf{B} = 0$; the general coalescing case was treated above.

4 Transformations and invariants

Here $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is a complex paravector: $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Transformation factors are constant complex paravectors. Distinguish the scalar part $\text{sc}(Z) = S = \frac{1}{2} \text{tr} Z$ from (3.7), the complex determinant $S^2 - \mathbf{V}^2$ from (3.8), and the positive norm $\text{sc}(Z^H Z)$ from (3.39). Scalar part and trace carry exactly the same information; we use sc below.

4.1 Similarity, determinant, and norm

For a complex [unit paravector](#) T , similarity means $Z' = T^{-1} Z T$. Scalar-part cyclicity gives

$$\text{sc}(Z') = \text{sc}(T^{-1} Z T) = \text{sc}(Z T T^{-1}) = \text{sc}(Z). \quad (4.1)$$

Thus trace is also preserved. Multiplicativity gives

$$\det(Z') = \det(T^{-1}) \det(Z) \det(T) = \det(Z). \quad (4.2)$$

The opposite similarity $Z' = T Z T^{-1}$ has the same invariants. For independent complex factors T, T' , the law $Z' = T' Z T$ instead gives

$$\det(Z') = \det(T') \det(T) \det(Z). \quad (4.3)$$

This preserves the determinant for every Z exactly when $\det(T') \det(T) = 1$. This alone does not preserve scalar part, norm, or the real paravector subspace.

If the factor T is unitary, then $T^H = T^{-1}$ by (3.25). For $Z' = T^H Z T$, cancellation and scalar-part cyclicity give

$$\begin{aligned} \|Z'\|^2 &= \text{sc}(T^H Z^H T T^H Z T) \\ &= \text{sc}(Z^H Z T T^H) = \|Z\|^2. \end{aligned} \quad (4.4)$$

The norm is also preserved by $Z' = T Z$, $Z' = Z T$, and $Z' = T Z T^H$. The two similarities additionally preserve scalar part and determinant; a one-sided unitary product need not.

4.1.1 Unit determinant is not Hermitian unitarity

The word *unit* in (3.24) means invertible; *unitary* means (3.25). A useful third condition is [determinant one](#). To see their difference, let $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$ be a real paravector increment, with $dt > \|d\mathbf{r}\|$, and define the positive real scalar $ds = \sqrt{dt^2 - d\mathbf{r}^2}$. Reserve U for the real proper velocity $U = dX/ds$. Since $dX^* dX = ds^2$,

$$U^* U = \frac{dX^* dX}{ds^2} = 1. \quad (4.5)$$

Thus $\det(U) = 1$ and $U^{-1} = U^*$: U is a unit paravector with $U U^{-1} = U^{-1} U = 1$. Its letter alone implies no algebraic property. The two conjugates are

$$\begin{aligned} U^H &= U = \frac{dt + d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}, \\ U^* &= \frac{dt - d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}. \end{aligned}$$

Hermitian conjugation leaves every real paravector fixed; sigma conjugation reverses its vector part. Therefore

$$U^H U = U^2 = \frac{dt^2 + d\mathbf{r}^2 + 2 dt d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds^2}, \quad (4.6)$$

$$\|U\|^2 = \frac{dt^2 + d\mathbf{r}^2}{dt^2 - d\mathbf{r}^2}. \quad (4.7)$$

Both expressions equal 1 only when $d\mathbf{r} = 0$, its rest frame. Section 6 develops the kinematics of this proper velocity.

4.2 Real paravectors and the Lorentz metric

Let $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ be real, with $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$. Equation (3.8) gives $\det(X) = a^2 - \mathbf{A}^2$, with Lorentz signature; (3.21) gives $\|X\|^2 = a^2 + \mathbf{A}^2$, with Euclidean signature. A transformation of real spacetime paravectors must preserve both reality and the determinant. For a complex [unit paravector](#) T with $\det(T) = 1$, these requirements hold for the Hermitian congruence

$$\boxed{X' = T^{-1} X (T^{-1})^H}. \quad (4.8)$$

Reality and determinant are preserved because

$$\begin{aligned} X'^H &= T^{-1} X^H (T^{-1})^H = X', \\ \det(X') &= \det(T^{-1}) \det(X) \det((T^{-1})^H) = \det(X). \end{aligned}$$

These congruences give the proper, future-preserving Lorentz transformations. Future X has nonnegative eigenvalues $a \pm \|\mathbf{A}\|$ and a real square root. Its image satisfies $\text{sc}(X') = \|T^{-1} \sqrt{X}\|^2 \geq 0$. Together with determinant invariance, this preserves the future cone. The factors T and $-T$ give the same transformation [1, Secs. II and IV].

For real θ and a [real unit vector](#) $\mathbf{u}$, the algebraic definitions $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ reduce this congruence to $X' = R^H X R$ and $X' = L^{-1} X L^{-1}$, respectively. Their components are derived in the next two subsections. For $X = 1$, the invariants differ explicitly:

$$\begin{aligned} X' &= L^{-2} = \cosh \theta - \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \det(X') &= 1, \\ \text{sc}(X') &= \cosh \theta, \\ \|X'\|^2 &= \cosh(2\theta). \end{aligned} \quad (4.9)$$

Thus the determinant supplies the spacetime metric, while the object being transformed determines whether to use congruence or similarity.

4.3 Rotation action and Rodrigues' formula

Recall the unitary factor $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.55), with real angle θ and [real unit axis](#) $\mathbf{u}$. For the complex paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$Z' = R^H Z R = R^{-1} Z R. \quad (4.10)$$

This keeps $S' = S$. The vector product (3.29) gives

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2(\mathbf{u} \cdot \mathbf{V}) \mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}, \\ [\mathbf{V} \cdot \boldsymbol{\sigma}, \mathbf{u} \cdot \boldsymbol{\sigma}] &= -2i(\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Inserting the sine and cosine forms of R and R^{-1} therefore yields

$$\begin{aligned} \mathbf{V}' \cdot \boldsymbol{\sigma} &= \cos^2 \frac{\theta}{2} \mathbf{V} \cdot \boldsymbol{\sigma} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} (\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma} \\ &\quad + \sin^2 \frac{\theta}{2} (2(\mathbf{u} \cdot \mathbf{V}) \mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Collecting terms gives Rodrigues' formula:

$$\boxed{\mathbf{V}' = \mathbf{V} \cos \theta + (\mathbf{u} \times \mathbf{V}) \sin \theta + (\mathbf{u} \cdot \mathbf{V})(1 - \cos \theta)\mathbf{u}.} \quad (4.11)$$

This is a right-handed rotation through $+\theta$ about $\mathbf{u}$. It preserves sc, determinant, norm, and the real paravector subspace. For complex $\mathbf{V}$, it rotates its real and imaginary parts by the same real rotation. The sign change in (3.67) affects both factors in (4.10), so Z' has period 2π .

4.4 Boost action and its sign convention

Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (3.56); $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. For a complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, choose the congruence

$$Z' = L^{-1} Z L^{-1}. \quad (4.12)$$

The factors have the same sign because $L^H = L$ by (3.62). Write the complex vector as the sum of its parallel and perpendicular parts:

$$\begin{aligned} \mathbf{V}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{V})\mathbf{u}, \\ \mathbf{V}_{\perp} &:= \mathbf{V} - \mathbf{V}_{\parallel}. \end{aligned}$$

The parallel term commutes with $\mathbf{u} \cdot \boldsymbol{\sigma}$; the perpendicular term anticommutes with it. Consequently,

$$\begin{aligned} L^{-1}(S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L^{-1} &= (S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L^{-2}, \\ L^{-1}(\mathbf{V}_{\perp} \cdot \boldsymbol{\sigma})L^{-1} &= \mathbf{V}_{\perp} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Using $L^{-2} = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma})$ from (3.72), multiplication gives

$$\boxed{S' = \gamma(S - \beta \mathbf{u} \cdot \mathbf{V}).} \quad (4.13)$$

$$\boxed{\mathbf{V}' = \mathbf{V}_{\perp} + \gamma(\mathbf{V}_{\parallel} - \beta S \mathbf{u}).} \quad (4.14)$$

These preserve $S^2 - \mathbf{V}^2$ and reality; they generally change S and $\|\mathbf{V}\|^2$.

The inverse signs are a convention: (3.58) gives $L^{-1} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = L(-\theta, \mathbf{u})$. Replacing θ by $-\theta$ changes (4.12) to $Z' = LZL$ and changes β to $-\beta$ in the component equations. The two conventions describe opposite boosts. We keep the positive exponential definition of L throughout; in the spacetime application, (4.12) gives components in the frame moving with velocity $+\beta \mathbf{u}$.

A boost similarity is a different action. For $Z' = L^{-1}ZL$, the same commutation rules instead give $S' = S$ and

$$\mathbf{V}' = \mathbf{V}_{\parallel} + \gamma(\mathbf{V}_{\perp} - i\beta \mathbf{u} \times \mathbf{V}). \quad (4.15)$$

This similarity preserves scalar part and determinant, but a real vector generally acquires an imaginary component. It will act on the complex electromagnetic field, whose real and imaginary vectors represent two physical fields.

5 Differentiated fields

5.1 Paravector differential operators

Let $t \in \mathbb{R}$ and $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$ be real coordinates. Define their scalar and vector derivatives by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}.$$

Define the partial paravector and its sigma conjugate by

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (5.1)$$

$$\partial^* = \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \quad (5.2)$$

The sigma generators are constant. Derivatives act from the left on everything to their right unless parentheses specify their scope. For fields with commuting second derivatives, anticommutation cancels all mixed spatial terms:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The mixed time-space terms likewise cancel. Both orders therefore give the scalar d'Alembertian

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} =: \square. \quad (5.3)$$

These are operator identities: $\partial \partial^* Z$ means $\partial(\partial^* Z)$.

5.2 Components and the product rule

Let the complex paravector field be

$$Z(t, \mathbf{r}) = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma},$$

with real scalar functions $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$ and real vector functions $\mathbf{A}, \mathbf{B} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. Ordered vector multiplication gives its four real components:

$$\begin{aligned} \partial Z &= (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \\ &\quad + (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\partial_{\mathbf{r}} b + \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (5.4)$$

Multiplication by i exchanges real and imaginary parts, with the former imaginary parts acquiring a minus sign.

For a second complex paravector field $Z'(t, \mathbf{r})$, the prime labels the field rather than a derivative. Apply the scalar product rule before reordering any factors:

$$\begin{aligned} \partial(ZZ') &= (\partial_t Z)Z' + Z(\partial_t Z') \\ &\quad + \sum_{n=1}^3 \sigma_n \left(\frac{\partial Z}{\partial r_n} Z' + Z \frac{\partial Z'}{\partial r_n} \right) \\ &= (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 [\sigma_n, Z] \frac{\partial Z'}{\partial r_n}. \end{aligned}$$

Define the last sum as $[\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z' := \sum_n [\sigma_n, Z] \partial Z' / \partial r_n$. Changing the spatial sign gives the paired rules

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z', \quad (5.5)$$

$$\partial^*(ZZ') = (\partial^* Z)Z' + Z(\partial^* Z') - [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (5.6)$$

The contraction retains the displayed multiplied order.

For a noncommuting check, take $Z = \sigma_2$ and $Z' = r_1$. Since $[\sigma_1, \sigma_2] = 2i\sigma_3$, the two right-hand sides give

$$\begin{aligned} \partial(\sigma_2 r_1) &= \sigma_2 \sigma_1 + [\sigma_1, \sigma_2] = -i\sigma_3 + 2i\sigma_3 = i\sigma_3, \\ \partial^*(\sigma_2 r_1) &= -\sigma_2 \sigma_1 - [\sigma_1, \sigma_2] = i\sigma_3 - 2i\sigma_3 = -i\sigma_3. \end{aligned}$$

These agree with direct differentiation, $\pm\sigma_1\sigma_2$. The ordinary Leibniz rule holds when the left factor commutes with every σ_n , in particular for a complex scalar. Otherwise the commutator term must be retained.

6 Space–time kinematics

Let $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ be a real event, with time $t \in \mathbb{R}$ and position $\mathbf{r} \in \mathbb{R}^3$. A massive particle follows a future-directed time-like path $\mathbf{r}(t)$; s denotes its proper time.

6.1 Intervals, velocity, and momentum

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. The differential event and its invariant interval are

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\ ds^2 &= \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2. \end{aligned} \quad (6.1)$$

By (4.8), a constant complex [unit paravector](#) T with $\det T = 1$ gives

$$dX' = T^{-1} dX (T^{-1})^H, \quad \det(dX') = ds^2.$$

Positive, zero, and negative values of (6.1) mean time-like, null, and space-like separation. Massive particles follow time-like paths; light follows null paths; causal signals cannot connect space-like separated events. For two real intervals,

$$\begin{aligned} [dX, dX'] &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}, \\ (dX)^{-1} &= \frac{dX^*}{ds^2} \quad (ds^2 \neq 0). \end{aligned}$$

A null interval has no inverse.

Along the massive path, define the real coordinate velocity $\mathbf{v} = d\mathbf{r}/dt$ and choose $ds > 0$. Then $ds^2 = (1 - \mathbf{v}^2) dt^2$ gives

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{dt}{ds}, \quad (6.2)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}. \quad (6.3)$$

Applying (6.3) with (6.2) gives the real proper velocity

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.4)$$

The normalization (4.5) gives $U^*U = 1$ and $U^{-1} = U^* = \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma})$. Since $U^H = U$, this is [determinant one](#), not Hermitian unitarity except at rest. For a [real unit direction](#) $\mathbf{u}$, write $\mathbf{v} = \beta\mathbf{u}$ and $\theta = \tanh^{-1}\beta$. Equation (3.72) gives $U = L^2$ for $L = \exp(\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$.

For constant real mass $m > 0$, define the real four-momentum

$$P := mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (6.5)$$

Its real coefficients are energy and momentum:

$$\begin{aligned} E &= \text{sc}(P) = m\gamma, \\ \mathbf{p} &= \text{vec}(P) = m\gamma\mathbf{v}, \\ P^{-1} &= \frac{E - \mathbf{p} \cdot \boldsymbol{\sigma}}{m^2}. \end{aligned}$$

Since $\det U = 1$, the determinant gives the mass shell

$$E^2 - \mathbf{p}^2 = \det P = m^2. \quad (6.6)$$

For a massive particle of finite energy,

$$\mathbf{v}^2 = \frac{\mathbf{p}^2}{E^2} = 1 - \frac{m^2}{E^2} < 1.$$

Approaching light speed requires unbounded energy. A massless path has no proper-time velocity. Define its momentum directly as a future null paravector $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with $E > 0$ and $E^2 = \mathbf{p}^2$; it has no inverse.

6.2 Proper acceleration

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$, and the real coordinate velocity $\mathbf{v}(t)$ and Lorentz factor γ . Define the real coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$. Differentiating gives

$$\begin{aligned} \frac{d\gamma}{dt} &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2}(-2\mathbf{v} \cdot \mathbf{a}) = \gamma^3\mathbf{v} \cdot \mathbf{a}, \\ \frac{d^2}{ds^2} &= \gamma \frac{d}{dt} \left(\gamma \frac{d}{dt} \right) = \gamma^2 \frac{d^2}{dt^2} + \gamma^4(\mathbf{v} \cdot \mathbf{a}) \frac{d}{dt}. \end{aligned}$$

Applying this to X yields the real proper-acceleration paravector:

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^2\mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.7)$$

Differentiating $U^*U = 1$ gives

$$\left(\frac{dU}{ds} \right)^* U + U^* \frac{dU}{ds} = 0.$$

The two terms have equal real scalar parts, so

$$\text{sc}(U^* dU/ds) = 0. \quad (6.8)$$

Thus four-velocity and four-acceleration are Minkowski-orthogonal. Multiplication on the right by U^* also gives

$$\begin{aligned} \text{adj}\left(\frac{dU}{ds}\right) &= -U^* \frac{dU}{ds} U^*. \\ U^* \frac{dU}{ds} &= \gamma^3 (\mathbf{a} - i\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (6.9)$$

The real vector records acceleration and the imaginary vector records turning of the velocity. Since $\mathbf{a} \cdot (\mathbf{v} \times \mathbf{a}) = 0$,

$$\begin{aligned} \det\left(\frac{dU}{ds}\right) &= -\left(U^* \frac{dU}{ds}\right)^2 \\ &= -\gamma^6 (\mathbf{a}^2 - (\mathbf{v} \times \mathbf{a})^2) \\ &= -\gamma^4 (\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2). \end{aligned} \quad (6.10)$$

The last equality uses $(\mathbf{v} \times \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}^2 - (\mathbf{v} \cdot \mathbf{a})^2$ and $1 - \mathbf{v}^2 = \gamma^{-2}$. Nonzero proper acceleration is therefore space-like; its invariant squared magnitude is the negative determinant. For $\mathbf{v} \neq 0$, define $\mathbf{a}_{\parallel} = (\mathbf{v} \cdot \mathbf{a})\mathbf{v}/v^2$ and $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$. Equivalently,

$$-\det\left(\frac{dU}{ds}\right) = \gamma^4 \mathbf{a}_{\perp}^2 + \gamma^6 \mathbf{a}_{\parallel}^2. \quad (6.11)$$

At rest this reduces to $\mathbf{a}^2$.

Energy and inertia. Recall $E = m\gamma$ and $\mathbf{p} = m\gamma\mathbf{v}$ for invariant real mass $m > 0$. Differentiating, using the derivative of γ above, gives

$$\frac{d\mathbf{p}}{dt} = m\gamma\mathbf{a} + m\gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.$$

For $\mathbf{v} \neq 0$, resolving along and perpendicular to $\mathbf{v}$ yields

$$\boxed{\frac{d\mathbf{p}}{dt} = m\gamma\mathbf{a}_{\perp} + m\gamma^3\mathbf{a}_{\parallel}.} \quad (6.12)$$

Taking its dot product with $\mathbf{v}$ gives the work rate

$$\frac{dE}{dt} = m\gamma^3 \mathbf{v} \cdot \mathbf{a} = \mathbf{v} \cdot \frac{d\mathbf{p}}{dt}. \quad (6.13)$$

At rest the force is $m\mathbf{a}$. Restoring c , $E = \gamma mc^2$; $E/c^2 = \gamma m$ is sometimes called *relativistic mass*. We retain invariant m : the longitudinal and transverse inertia in (6.12) differ [2, pp. 136–137, 140].

6.3 World lines, clocks, and light cones

6.3.1 Integrating proper velocity

Recall the real path $X(s) = t(s) + \mathbf{r}(s) \cdot \boldsymbol{\sigma}$ and its tangent $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. For real proper times $s_0 < s_1$, integration gives

$$\boxed{X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds.} \quad (6.14)$$

Its real components are

$$\begin{aligned} t(s_1) - t(s_0) &= \int_{s_0}^{s_1} \gamma(s) ds, \\ \mathbf{r}(s_1) - \mathbf{r}(s_0) &= \int_{s_0}^{s_1} \gamma(s) \mathbf{v}(s) ds. \end{aligned}$$

Thus between coordinate times t_0 and t the clock records

$$\boxed{s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'.} \quad (6.15)$$

This is the standard world-line proper-time integral [2, pp. 133–134, eq. (7.19)]. For constant U , (6.14) is the straight line $X(s) = X(s_0) + (s - s_0)U$. For varying U , integrate the tangent; an endpoint boost alone does not specify the path.

6.3.2 Clock comparison

Let $X_0 = X(s_0)$ and $X_1 = X(s_1)$ be fixed, time-like separated events. Write $X_k = t_k + \mathbf{r}_k \cdot \boldsymbol{\sigma}$ for $k = 0, 1$. In their separation's rest frame, $\mathbf{r}_1 = \mathbf{r}_0$ and $t_1 - t_0 = \sqrt{\det(X_1 - X_0)}$. Since $\sqrt{1 - \mathbf{v}^2} \leq 1$, (6.15) gives

$$s_1 - s_0 \leq \sqrt{\det(X_1 - X_0)}. \quad (6.16)$$

Equality requires the straight inertial path. The determinant of a finite displacement therefore does not generally equal the squared elapsed proper time of an accelerated clock.

For a return trip lasting real coordinate time $T > 0$, take velocities $+\beta \mathbf{u}$ and $-\beta \mathbf{u}$ for $T/2$ each, with [real unit vector](#) $\mathbf{u}$ and $0 < \beta < 1$. The ideal instantaneous reversal is the limit of a short smooth turn. The traveling clock records

$$s(T) - s(0) = T\sqrt{1 - \beta^2},$$

while the clock at rest records T . This is the twin-clock comparison [2, pp. 120–122].

6.3.3 Constant proper acceleration

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. Let $a > 0$ be a constant real proper-acceleration magnitude and $\mathbf{u}$ a fixed [real unit axis](#). Starting at $X(0) = 0$ and $U(0) = 1$, take rapidity $\theta(s) = as$. Then

$$U(s) = L(as, \mathbf{u})^2 = \cosh(as) + \sinh(as)\mathbf{u} \cdot \boldsymbol{\sigma}.$$

Differentiation verifies the stated acceleration:

$$-\det\left(\frac{dU}{ds}\right) = a^2(\cosh^2(as) - \sinh^2(as)) = a^2.$$

Integrating (6.14) yields

$$\boxed{X(s) = \frac{\sinh(as)}{a} + \frac{\cosh(as) - 1}{a} \mathbf{u} \cdot \boldsymbol{\sigma}.} \quad (6.17)$$

Its components give

$$\begin{aligned} \mathbf{v}(s) &= \frac{d\mathbf{r}/ds}{dt/ds} = \tanh(as)\mathbf{u}, \\ (1 + a\mathbf{u} \cdot \mathbf{r})^2 - (at)^2 &= 1. \end{aligned}$$

Thus (6.17) is the usual hyperbolic motion, with speed below light speed at every finite s [2, pp. 142–143].

6.3.4 Light cones

For a real event separation $\Delta X = \Delta t + \Delta \mathbf{r} \cdot \boldsymbol{\sigma}$, the eigenvalues are $\Delta t \pm \|\Delta \mathbf{r}\|$. Both are nonnegative precisely in the future causal cone:

$$\Delta t \geq \|\Delta \mathbf{r}\|. \quad (6.18)$$

The boundary has $\det(\Delta X) = 0$; its nonzero future rays are

$$\Delta X = \Delta t(1 + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Delta t > 0, \quad \mathbf{u}^2 = 1.$$

The Hermitian congruence (4.8) preserves nonnegativity of the eigenvalues and the determinant, hence the cone and its boundary. For a future time-like or null path,

$$\|\Delta \mathbf{r}\| = \left\| \int \mathbf{v} dt \right\| \leq \int \|\mathbf{v}\| dt \leq \Delta t.$$

Its endpoints therefore satisfy (6.18); a space-like separation cannot carry a causal signal [2, pp. 117–118].

6.4 Free-particle action

Recall real $m > 0$, $\mathbf{v} = d\mathbf{r}/dt$, and $ds = dt/\gamma$. Here $L(t, \mathbf{r}, \mathbf{v})$ is a real scalar Lagrangian, not the boost factor. The free action is the physical choice

$$\int L dt = -m \int ds. \quad (6.19)$$

Using $ds = dt/\gamma$ gives the Lagrangian

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m/\gamma. \quad (6.20)$$

For endpoint-fixed real path variations $\delta \mathbf{r}(t)$, varying the Lagrangian and integrating by parts gives

$$\delta \int L dt = \int \left(\partial_{\mathbf{r}} L - \frac{d}{dt} \partial_{\mathbf{v}} L \right) \cdot \delta \mathbf{r} dt.$$

Stationarity for every variation yields the Euler–Lagrange equation. With real Hamiltonian $H := \mathbf{v} \cdot \partial_{\mathbf{v}} L - L$, the chain rule gives

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L, \quad (6.21)$$

$$\frac{dH}{dt} = \mathbf{v} \cdot \left(\frac{d}{dt} \partial_{\mathbf{v}} L - \partial_{\mathbf{r}} L \right) - \partial_t L = -\partial_t L. \quad (6.22)$$

By (6.21) and (6.22), spatial and temporal translation invariance conserve canonical momentum and energy. For the free action,

$$\begin{aligned} \partial_{\mathbf{v}} L &= m\gamma \mathbf{v} = \mathbf{p}, \\ H &= m\gamma \mathbf{v}^2 + m/\gamma = m\gamma = E. \end{aligned}$$

Writing $v = \|\mathbf{v}\| \ll 1$, the binomial series gives

$$\begin{aligned} \sqrt{1 - v^2} &= 1 - \frac{1}{2}v^2 + O(v^4), \\ \gamma &= 1 + \frac{1}{2}v^2 + O(v^4), \\ L &= -m + \frac{1}{2}mv^2 + O(mv^4), \\ H &= m + \frac{1}{2}mv^2 + O(mv^4). \end{aligned}$$

The exact kinetic energy above rest is

$$K := H - m = m(\gamma - 1). \quad (6.23)$$

This and both expansions recover $mv^2/2$ at low speed after removal of the constant rest term.

7 Electromagnetism

All coefficient fields in this section are real functions of real $(t, \mathbf{r})$. Define

$$\begin{aligned} F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} && \text{electric and magnetic fields,} \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma} && \text{scalar and vector potentials,} \\ D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} && \text{charge and current densities.} \end{aligned} \quad (7.1)$$

Thus Φ, D are real paravectors, while F is a complex pure vector. Recall $\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}$ and $\partial^* \partial = \square = \partial_t^2 - \partial_{\mathbf{r}}^2$.

The dot/cross product (3.29) combines metric and orientation. With a spatial derivative it gives divergence and curl, while the six real components of F represent an antisymmetric spacetime field tensor. This algebra of physical space is isomorphic to the even subalgebra of spacetime algebra; its real paravectors represent spacetime vectors relative to an observer [1, Secs. IV and V]. The product explains the compact form; Maxwell dynamics remains physical input, derived from an action in Section 7.3.

7.1 Maxwell's equations and wave propagation

Maxwell's four equations are the single field law

$$\boxed{\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.} \quad (7.2)$$

The product rule expands its left side into

$$\begin{aligned}\partial F = & \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i\partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \\ & + \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \\ & + i \underbrace{(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}.\end{aligned}$$

Equating the four parts gives

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (7.3)$$

$$\partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (7.4)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}, \quad (7.5)$$

$$\partial_{\mathbf{r}} \times \mathbf{E} + \partial_t \mathbf{B} = 0. \quad (7.6)$$

Equations (7.3)–(7.6) are Gauss’ electric and magnetic laws, Ampère–Maxwell’s law, and Faraday’s law [3, p. 6]. Applying ∂^* to (7.2) gives

$$\begin{aligned}\square F &= \partial^* D^* = (\partial D)^* \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i\partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned} \quad (7.7)$$

The scalar part vanishes because F is scalar-free. Thus

$$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (7.8)$$

$$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (7.9)$$

$$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (7.10)$$

Equation (7.8) is charge conservation; (7.9) and (7.10) are the sourced field waves. In vacuum, $D = 0$ and both fields obey the free wave equation.

7.2 Potentials and gauge freedom

Recall the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. Its derivative is

$$\partial \Phi = (\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}) + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma} + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.$$

Define the real scalar $S := \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. Identifying the remaining components with F^* gives

$$\partial \Phi = S + F^*, \quad (7.11)$$

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad (7.12)$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (7.13)$$

Equation (7.11) separates the gauge scalar from the field; (7.12) and (7.13) satisfy (7.4) and (7.6) identically. Since $F = \partial^* \Phi^* - S$,

$$\begin{aligned}\partial F &= \square \Phi^* - \partial S \\ &= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Comparison with (7.2) yields

$$\rho = \square V - \partial_t S, \quad (7.14)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (7.15)$$

These source equations hold before gauge fixing. For a real scalar gauge function $\lambda(t, \mathbf{r})$, let

$$\Phi' = \Phi + \partial^* \lambda. \quad (7.16)$$

Differentiating gives $\partial \Phi' = \partial \Phi + \square \lambda$. Only the scalar changes: $S' = S + \square \lambda$, while $F' = F$. In the Lorenz gauge $S = 0$, the source equations reduce to $\square V = \rho$ and $\square \mathbf{A} = \mathbf{J}$.

7.3 The electromagnetic field action

For real scalar potential $V(t, \mathbf{r})$ and vector potential $\mathbf{A}(t, \mathbf{r})$, write $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. With prescribed D , Maxwell dynamics follows from the real Lagrangian density

$$L = \tfrac{1}{2} \text{re}(F^2) - \text{sc}(\Phi^* D) = \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho V + \mathbf{J} \cdot \mathbf{A}. \quad (7.17)$$

In varying the action, boundary-vanishing variations satisfy

$$\begin{aligned} \delta \mathbf{E} &= -\partial_{\mathbf{r}} \delta V - \partial_t \delta \mathbf{A}, \\ \delta \mathbf{B} &= \partial_{\mathbf{r}} \times \delta \mathbf{A}. \end{aligned}$$

Integration by parts gives

$$\delta \int L \, dt \, d^3 \mathbf{r} = \int ((\partial_{\mathbf{r}} \cdot \mathbf{E} - \rho) \delta V + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} + \mathbf{J}) \cdot \delta \mathbf{A}) \, dt \, d^3 \mathbf{r}. \quad (7.18)$$

Independent variations give (7.3) and (7.5); the potential identities supply (7.4) and (7.6). Together they recover (7.2). The symbol δ denotes variation.

7.4 Field invariants

Since $\text{sc}(F) = 0$, its adjugate is $-F$. Squaring gives

$$F^2 = -\det F = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}. \quad (7.19)$$

For a complex [unit paravector](#) T , the field similarity $F' = T^{-1} F T$ preserves this scalar by (4.2). Physical rotations and boosts use $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, respectively, with [real unit vector](#) $\mathbf{u}$ and real angle or rapidity θ ; their field laws are derived in Section 11. The real and imaginary parts of (7.19) are therefore observer-independent. For a nonzero field, they give the classification

Condition	Consequence
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 > \mathbf{B}^2$	Some frame has $\mathbf{B}' = 0$.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 < \mathbf{B}^2$	Some frame has $\mathbf{E}' = 0$.
$\mathbf{E} \cdot \mathbf{B} \neq 0$	Neither field can be eliminated.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 = \mathbf{B}^2$	Null field: equal, perpendicular fields.

7.5 Lorentz force from the action

Recall the real proper velocity $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$, a [unit paravector](#) with $UU^{-1} = 1$. Let $q \in \mathbb{R}$ be charge and $m > 0$ constant mass. The real momentum is $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with $E = m\gamma$ and $\mathbf{p} = m\gamma\mathbf{v}$. The real particle Lagrangian L is fixed by the action

$$\int L dt = -m \int ds - q \int \text{sc}(\Phi^* U) ds. \quad (7.20)$$

Since $\text{sc}(\Phi^* U) = \gamma(V - \mathbf{v} \cdot \mathbf{A})$, the Lagrangian is

$$L = -m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}. \quad (7.21)$$

Its Euler–Lagrange terms are

$$\begin{aligned} \partial_{\mathbf{v}} L &= \mathbf{p} + q\mathbf{A}, \\ \frac{d}{dt} \partial_{\mathbf{v}} L &= \frac{d\mathbf{p}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \\ \partial_{\mathbf{r}} L &= -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Here spatial differentiation holds the independent velocity argument fixed. Using $\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.22)$$

Here (7.12) and (7.13) turn the potential derivatives into the fields. The Hamiltonian and its balance law are

$$H = \mathbf{v} \cdot (\mathbf{p} + q\mathbf{A}) - L = E + qV. \quad (7.23)$$

Using (6.22) in (7.23),

$$\begin{aligned} \frac{dE}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V &= -\partial_t L = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}. \\ \frac{dE}{dt} &= q\mathbf{v} \cdot \mathbf{E}. \end{aligned} \quad (7.24)$$

This is the mechanical work rate [3, pp. 115–116]. Thus $\mathbf{p} + q\mathbf{A}$ and H are canonical momentum and energy; $\mathbf{p}, E$ are mechanical momentum and energy, with E including rest energy. Canonical energy is conserved for static potentials in the chosen gauge. A time-dependent gauge transformation can change H while leaving E and the force unchanged.

Now expand the complex product of the field and proper velocity:

$$\begin{aligned} \frac{FU}{\gamma} &= \mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} - \mathbf{v} \times \mathbf{E}) \cdot \boldsymbol{\sigma}). \end{aligned} \quad (7.25)$$

The real scalar and vector parts reproduce (7.24) and (7.22). Since P is real and $d/ds = \gamma d/dt$, they combine into

$$\boxed{\frac{dP}{ds} = q \text{re}(FU) = \frac{q}{2}(FU + UF^H).} \quad (7.26)$$

Hermitian extraction enforces real mechanical momentum and reproduces the action-derived force. The imaginary part is the magnetic-dual force in Section 13.1; it need not vanish for an electric charge.

7.6 Acceleration and the mass shell

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$ and $U^{-1} = U^*$, real momentum $P = mU$, and coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$. Scalar-part cyclicity proves tangency to the mass shell:

$$\begin{aligned} \text{sc}(U^* dP/ds) &= \frac{q}{2} \text{sc}(U^* F U + F^H) \\ &= \frac{q}{2} (\text{sc}(F) + \text{sc}(F^H)) = 0. \end{aligned}$$

This agrees with $\text{sc}(U^* dU/ds) = 0$ in Section 6.2. Combining (6.7) with the force components in (7.25) gives

$$\begin{aligned} m\gamma^3 \mathbf{v} \cdot \mathbf{a} &= q \mathbf{v} \cdot \mathbf{E}, \\ m\gamma (\mathbf{a} + \gamma^2 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v}) &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

Eliminating $\mathbf{v} \cdot \mathbf{a}$ yields the coordinate acceleration

$$\boxed{\mathbf{a} = \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E}) \mathbf{v})}. \quad (7.27)$$

The magnetic term changes momentum without doing work. Taking the determinant of the real force gives

$$\begin{aligned} -\det\left(\frac{dP}{ds}\right) &= -m^2 \det\left(\frac{dU}{ds}\right) \\ &= q^2 \gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \end{aligned} \quad (7.28)$$

Dividing by m^2 gives the invariant squared proper acceleration. The factor $\gamma^2((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2)$ is the squared electric field in the particle's instantaneous rest frame.

7.7 Energy and momentum flux

Recall $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Its Hermitian product is

$$\frac{1}{2} F F^H = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (7.29)$$

Define the real energy density $u = (\mathbf{E}^2 + \mathbf{B}^2)/2$ and real Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}$. These coefficients are frame-dependent, whereas F^2 contains the Lorentz invariants. The full stress–energy tensor is represented by the real linear map

$$T(N) := \frac{1}{2} F N F^H, \quad (7.30)$$

with real paravector input N . Its time-direction value is $T(1) = u + \mathbf{S} \cdot \boldsymbol{\sigma}$. It is future causal because $u \geq 0$ and

$$\det(T(1)) = u^2 - \mathbf{S}^2 = \frac{1}{4} \|\det F\|^2 \geq 0. \quad (7.31)$$

Equality holds for null fields. Reversing the order gives $F^H F = (F F^H)^*$: the same energy density and opposite flux. The transformation of the full map is derived in Section 11.

Using Maxwell's vector equations,

$$\begin{aligned} \partial_t u &= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \\ &= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \end{aligned}$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Thus Poynting's energy-balance law is

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.32)$$

Equivalently, the left side is $\frac{1}{2} \partial_t \text{sc}(FF^{\mathbf{H}}) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(FF^{\mathbf{H}})$. The right side is the field's loss of energy to matter per unit volume.

8 Relativistic wave equations

Here $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ has real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$, $m > 0$ is mass, $q \in \mathbb{R}$ is charge, and $\hbar = c = 1$. Amplitudes are complex functions of these real coordinates; their scalar or paravector type is specified below.

8.1 Amplitudes and their pairing

For complex paravectors ψ, ψ' and a left-acting paravector Z , define the complex scalar pairings

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^{\mathbf{H}} \psi), \quad (8.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^{\mathbf{H}} Z \psi). \quad (8.2)$$

Their real norm density is

$$\rho := \text{sc}(\psi^{\mathbf{H}} \psi) = \|\psi\|^2 \geq 0. \quad (8.3)$$

For a normalized Pauli wavefunction, $\int \rho \, d^3 \mathbf{r} = 1$; Dirac uses the sum over its two entries. This probability interpretation does not apply to an arbitrary scalar KG field.

For $\rho > 0$ and a real observable $Z^{\mathbf{H}} = Z$, its mean is

$$\langle Z \rangle_{\psi} := \frac{\text{sc}(\psi^{\mathbf{H}} Z \psi)}{\rho}. \quad (8.4)$$

For a spatial wavefunction, integrate numerator and denominator. For differential observables, derivatives act to the right and boundary terms must vanish when taking the Hermitian conjugate.

For arbitrary constant perpendicular **real unit vectors** $\mathbf{u}, \mathbf{a}$, choose a constant complex paravector ψ_+ of unit norm with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_+ = \psi_+$, and set $\psi_- = (\mathbf{a} \cdot \boldsymbol{\sigma})\psi_+$. Then

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_{\pm} = \pm \psi_{\pm}. \quad (8.5)$$

The pair is orthonormal: $\langle \psi_a | \psi_b \rangle = \delta_{ab}$ for $a, b = +, -$. A Pauli amplitude or either Dirac entry is

$$\psi(t, \mathbf{r}) = c_+(t, \mathbf{r})\psi_+ + c_-(t, \mathbf{r})\psi_-. \quad (8.6)$$

Here $c_+, c_- : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\rho = \|c_+\|^2 + \|c_-\|^2$. Both Dirac entries use the same fixed pair. Its construction and matrix verification are in Appendix A.4.

8.2 Energy and momentum as differential operators

Recall $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$. Temporarily restore Planck's constant $h > 0$ and $\hbar = h/(2\pi)$, keeping $c = 1$. For a free wave, let $\omega \in \mathbb{R}$ be angular frequency (2π times cycles per unit time), and let $\mathbf{k} \in \mathbb{R}^3$ be its wavenumber vector in cycles per unit length. For $\mathbf{k} \neq 0$, the wavelength $\lambda > 0$ satisfies $\|\mathbf{k}\| = 1/\lambda$.

The physical input is the Planck–Einstein energy relation and de Broglie momentum relation [4, pp. 248–249]:

$$E = \hbar\omega, \quad (8.7)$$

$$\mathbf{p} = \hbar\mathbf{k} = 2\pi\hbar\mathbf{k}. \quad (8.8)$$

Here $E \in \mathbb{R}$ is total energy, including rest energy, and $\mathbf{p} \in \mathbb{R}^3$ is momentum; its magnitude is h/λ . Photoelectric measurements test the energy relation [5, pp. 61–62]; electron diffraction tests the momentum relation [4, p. 255].

For the constant real four-momentum $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, the complex scalar plane wave is

$$\begin{aligned} f(X, P) &:= \exp(-i(\omega t - 2\pi\mathbf{k} \cdot \mathbf{r})) \\ &= \exp\left(-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{r})\right). \end{aligned}$$

The second line substitutes (8.7) and (8.8). Differentiating gives

$$\begin{aligned} i\hbar\partial_t f &= E f, \\ -i\hbar\partial_{\mathbf{r}} f &= \mathbf{p} f. \end{aligned}$$

Thus $i\hbar\partial_t$ and $-i\hbar\partial_{\mathbf{r}}$ represent energy and momentum on each mode, and extend to superpositions by linearity. Returning to $\hbar = 1$ means $h = 2\pi$, giving

$$f(X, P) = \exp(-i(Et - \mathbf{p} \cdot \mathbf{r})). \quad (8.9)$$

The two eigenvalue equations combine into

$$i\partial^* f = P f. \quad (8.10)$$

In a field, recall the real potential paravector $\Phi(t, \mathbf{r}) = V(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$: V is a real scalar potential and $\mathbf{A}$ a real vector potential. The real charge is q . The particle action gives canonical energy $E + qV$ and momentum $\mathbf{p} + q\mathbf{A}$; identifying them with translation operators defines the gauge-covariant mechanical operators

$$E := i\partial_t - qV, \quad (8.11)$$

$$\mathbf{p} := -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (8.12)$$

Here E includes rest energy. Below, $E, \mathbf{p}$ denote these differential operators unless real mode eigenvalues are specified. They act to the right and combine as

$$E - \mathbf{p} \cdot \boldsymbol{\sigma} = i\partial - q\Phi^*, \quad (8.13)$$

$$E + \mathbf{p} \cdot \boldsymbol{\sigma} = i\partial^* - q\Phi. \quad (8.14)$$

Operator order matters in a field; an ordinary paravector determinant cannot replace their product. For free commuting derivatives, $\square = \partial_t^2 - \partial_{\mathbf{r}}^2$ gives

$$(i\partial)(i\partial^*) = -\square. \quad (8.15)$$

Reversing the order gives the same result.

8.2.1 Field strength from noncommuting derivatives

Recall the real electric and magnetic fields from (7.12) and (7.13), and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real scalar in (7.11). The real potential is $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$; $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex vector paravector.

Apply the operators to a differentiable complex amplitude. The product rule cancels all derivatives of the amplitude in the commutators; for Cartesian indices $j, k = 1, 2, 3$,

$$\begin{aligned} [E, p_j] &= -iq(\partial_t A_j + \partial_j V) = iqE_j, \\ [p_j, p_k] &= iq(\partial_j A_k - \partial_k A_j), \\ \mathbf{p} \times \mathbf{p} &= iq\mathbf{B}. \end{aligned}$$

Sigma multiplication and the energy commutator consequently give

$$(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (8.16)$$

$$[E, \mathbf{p} \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}. \quad (8.17)$$

Here E_j is an electric-field component. Using (8.16) and (8.17) in the ordered products gives

$$\begin{aligned} (E + \mathbf{p} \cdot \boldsymbol{\sigma})(E - \mathbf{p} \cdot \boldsymbol{\sigma}) &= E^2 - (\mathbf{p} \cdot \boldsymbol{\sigma})^2 - [E, \mathbf{p} \cdot \boldsymbol{\sigma}] \\ &= E^2 - \mathbf{p}^2 - iqF, \end{aligned} \quad (8.18)$$

$$\begin{aligned} (E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma}) &= E^2 - (\mathbf{p} \cdot \boldsymbol{\sigma})^2 + [E, \mathbf{p} \cdot \boldsymbol{\sigma}] \\ &= E^2 - \mathbf{p}^2 - iqF^*. \end{aligned} \quad (8.19)$$

Their common scalar part gives KG; the vector terms give the Dirac spin coupling.

8.3 Klein–Gordon equation

Let $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ be a scalar field and $m > 0$ its real mass. For a free real four-momentum, $\det P = m^2$ means $E^2 - \mathbf{p}^2 = m^2$. Replacing its coefficients by the translation operators in (8.10) gives

$$((i\partial)(i\partial^*) - m^2)\psi = 0.$$

Using (8.15), this is the Klein–Gordon equation [6, p. 9]:

$$\boxed{(\square + m^2)\psi = 0.} \quad (8.20)$$

8.3.1 Positive and negative frequency

Recall the complex scalar field $\psi(t, \mathbf{r})$ and real mass $m > 0$. Define the positive scalar operator

$$\Omega := \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \quad \Omega^2 = \det(m + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}).$$

Since Ω commutes with ∂_t , KG factors as

$$\square + m^2 = \partial_t^2 + \Omega^2 = (\partial_t - i\Omega)(\partial_t + i\Omega). \quad (8.21)$$

The reverse product is the same. Its two frequency branches obey

$$(\partial_t \pm i\Omega)\psi_{\pm} = 0, \quad \psi_{\pm}(t, \mathbf{r}) = e^{\mp i\Omega t} \psi_{\pm}(0, \mathbf{r}).$$

Their sum gives the solution for arbitrary complex initial branch fields $\psi_{\pm}(0, \mathbf{r})$:

$$\boxed{\psi(t, \mathbf{r}) = e^{-i\Omega t} \psi_+(0, \mathbf{r}) + e^{i\Omega t} \psi_-(0, \mathbf{r}).} \quad (8.22)$$

8.3.2 Square roots from Fourier series

For spatially periodic fields, write the discrete Fourier series

$$\psi = \sum_n c_n f_n, \quad f_n(t, \mathbf{r}) = \exp(i(\omega_n t + 2\pi \mathbf{k}_n \cdot \mathbf{r})).$$

Here n labels modes, $c_n \in \mathbb{C}$ are constant coefficients, $\omega_n \in \mathbb{R}$ are signed angular phase frequencies, and $\mathbf{k}_n \in \mathbb{R}^3$ are wavenumber vectors in cycles per unit length. For $\mathbf{k}_n \neq 0$, its real wavelength $\lambda_n > 0$ satisfies $\|\mathbf{k}_n\| = 1/\lambda_n$. By (8.8), the mode momentum is $2\pi \mathbf{k}_n$ in our units. The $+\omega_n t$ convention gives energy eigenvalue $-\omega_n$. Differentiation gives

$$\begin{aligned} \square \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2) \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n. \end{aligned}$$

Thus each nonzero KG coefficient requires

$$\omega_n^2 = 4\pi^2 \mathbf{k}_n^2 + m^2, \quad \omega_n = \pm \sqrt{4\pi^2 \mathbf{k}_n^2 + m^2}.$$

Both signs are independent modes [6, pp. 22–23]; no common time period is assumed. A square-root operator acts on the same coefficients by

$$\sqrt{\square} \psi = \sum_n \sqrt{4\pi^2 \mathbf{k}_n^2 - \omega_n^2} c_n f_n.$$

Choose one square-root branch consistently. Squaring restores $\square$, giving

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square}). \quad (8.23)$$

For the positive spatial root, the same series gives

$$\begin{aligned} \Omega^2 \psi &= \sum_n (m^2 + 4\pi^2 \mathbf{k}_n^2) c_n f_n, \\ \Omega \psi &= \sum_n \sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} c_n f_n. \end{aligned}$$

Thus $e^{\mp i\Omega t}$ multiplies each spatial mode by $e^{\mp i\sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} t}$. Unlike Ω , $\sqrt{\square}$ has imaginary eigenvalues on KG modes; its branch does not select the frequency sign.

8.3.3 Potential

Recall the real potentials $V, \mathbf{A}$, charge q , and kinetic operators $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The scalar equation in a field is

$$\boxed{(-E^2 + \mathbf{p}^2 + m^2)\psi = 0.} \quad (8.24)$$

The mass-shell equation uses the scalar part $E^2 - \mathbf{p}^2$ of (8.18) or (8.19). The product rule gives

$$E^2 = -\partial_t^2 - iq(\partial_t V) - 2iqV\partial_t + q^2 V^2, \quad (8.25)$$

$$\mathbf{p}^2 = -\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2 \mathbf{A}^2. \quad (8.26)$$

Substituting (8.25) and (8.26) into (8.24) gives

$$\boxed{(\square + iqS - q^2 \det \Phi + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\psi = 0.} \quad (8.27)$$

All potential derivatives are retained [6, p. 139].

8.4 Dirac equation

8.4.1 Block form and free factorization

Recall $m > 0$ and real space–time coordinates $(t, \mathbf{r})$. A Dirac field is the two-entry block

$$\Psi(t, \mathbf{r}) = \begin{Bmatrix} \Psi_1(t, \mathbf{r}) \\ \Psi_2(t, \mathbf{r}) \end{Bmatrix}. \quad (8.28)$$

Each entry is a complex paravector amplitude of (8.6); the block therefore has four independent complex components. On blocks, Hermitian conjugation also transposes entries; sigma conjugation acts entrywise. The local pairing is

$$\langle \Psi' | \Psi \rangle := \sum_{k=1}^2 \text{sc}(\Psi_k'^H \Psi_k). \quad (8.29)$$

This gives the positive real density

$$\|\Psi\|^2 = \text{sc}(\Psi_1^H \Psi_1) + \text{sc}(\Psi_2^H \Psi_2). \quad (8.30)$$

For a complex paravector Z , define the complex-linear block map

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix}. \quad (8.31)$$

Here $\mathbf{I}$ is the two-entry block identity. Applying $\mathbf{W}(Z)$ twice multiplies the first entry by $Z \text{adj}(Z)$ and the second by $\text{adj}(Z) Z$. Both products are the paravector determinant, so

$$\mathbf{W}(Z)^2 = \det(Z) \mathbf{I}. \quad (8.32)$$

Sigma conjugation on each block entry gives $\mathbf{W}(Z)^* = \mathbf{W}(Z^*)$; Hermitian conjugation also transposes the block:

$$\mathbf{W}(Z)^H = \mathbf{W}(Z^*). \quad (8.33)$$

Thus $\mathbf{W}(Z)$ is not generally Hermitian. For the free field define

$$\begin{aligned} \mathbf{D} &:= \mathbf{W}(i\partial) - m\mathbf{I}, \\ \mathbf{D}' &:= \mathbf{W}(i\partial) + m\mathbf{I}. \end{aligned}$$

The Dirac equation is [6, p. 90]

$$\boxed{\mathbf{D}\Psi = 0}. \quad (8.34)$$

Since $\mathbf{W}(i\partial)^2 = -\square \mathbf{I}$, its mass conjugate gives

$$\mathbf{D}'\mathbf{D} = \mathbf{W}(i\partial)^2 - m^2 \mathbf{I} = -(\square + m^2) \mathbf{I}. \quad (8.35)$$

Applying (8.35) to (8.34) gives $(\square + m^2)\Psi = 0$: every Dirac component satisfies KG. The original first-order constraint is the coupled pair

$$i\partial\Psi_2 = m\Psi_1, \quad (8.36)$$

$$i\partial^*\Psi_1 = m\Psi_2. \quad (8.37)$$

8.4.2 Reconstructing free solutions from Klein–Gordon

Let $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$ solve KG and recall the constant basis amplitude ψ_+ . For constant complex scalars a, b , a two-constant construction is

$$\Psi = \frac{1}{m} \left\{ \begin{matrix} ma + ib\partial \\ ia\partial^* + mb \end{matrix} \right\} \psi' \psi_+. \quad (8.38)$$

Substituting (8.38) into (8.36) and (8.37) gives

$$\begin{aligned} i\partial\Psi_2 - m\Psi_1 &= -\frac{a}{m}(\square + m^2)\psi'\psi_+ = 0, \\ i\partial^*\Psi_1 - m\Psi_2 &= -\frac{b}{m}(\square + m^2)\psi'\psi_+ = 0. \end{aligned}$$

Taking $a = 1, b = 0$ gives one scalar-seeded solution. To span every Dirac field, instead let ψ have the form (8.6), with both scalar coefficients solving KG. Then

$$\Psi = \left\{ \begin{matrix} \psi \\ i\partial^*\psi/m \end{matrix} \right\}. \quad (8.39)$$

The second Dirac equation holds by construction; the first is KG. Conversely, every massive free solution has this form with $\psi = \Psi_1$. Equation (8.22) evolves its two coefficients.

On one Fourier mode, let $s \in \mathbb{C}$ satisfy $\mathbf{D}\Psi = s\Psi$ for a nonzero amplitude. Here $\square$ denotes its scalar mode eigenvalue. Since $\mathbf{D} + m\mathbf{I} = \mathbf{W}(i\partial)$, (8.35) gives

$$(\mathbf{D} + m\mathbf{I})^2\Psi = -\square\Psi \implies ((s + m)^2 + \square)\Psi = 0.$$

Thus the eigenvalues are

$$s = -m \pm i\sqrt{\square}.$$

When distinct, each allows two independent complex coefficients in Ψ_1 , with $\Psi_2 = i\partial^*\Psi_1/(s + m)$. Each root therefore occurs twice, giving the scalar characteristic equation

$$((s + m)^2 + \square)^2 = 0. \quad (8.40)$$

At $\square = 0$ the roots coalesce with total algebraic multiplicity four. For KG modes, $\square = -m^2$, so $s = 0, -2m$; the Dirac equation selects $s = 0$. The matrix verification is (A.18).

8.4.3 Potential and paravector components

Recall the real $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, real charge q , and $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. Minimal coupling gives

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (8.41)$$

Its entries are

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_2 = m\Psi_1, \quad (8.42)$$

$$(E + \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_1 = m\Psi_2. \quad (8.43)$$

Isolating the time derivative gives $i\partial_t \Psi = H\Psi$, where the Hamiltonian is

$$H := qV\mathbf{I} + \mathbf{W}(1)(m\mathbf{I} + \mathbf{W}(\mathbf{p} \cdot \boldsymbol{\sigma})). \quad (8.44)$$

Real potentials and vanishing boundary terms make H Hermitian in the integrated pairing, so it conserves the norm (8.30).

To expose each coefficient, write $\Psi_k = a_k + \mathbf{b}_k \cdot \boldsymbol{\sigma}$ for $k = 1, 2$, with complex scalar functions $a_k : \mathbb{R}^4 \rightarrow \mathbb{C}$ and complex vector functions $\mathbf{b}_k : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Sigma multiplication gives

$$Ea_2 - \mathbf{p} \cdot \mathbf{b}_2 = ma_1, \quad (8.45)$$

$$E\mathbf{b}_2 - \mathbf{p}a_2 - i\mathbf{p} \times \mathbf{b}_2 = m\mathbf{b}_1, \quad (8.46)$$

$$Ea_1 + \mathbf{p} \cdot \mathbf{b}_1 = ma_2, \quad (8.47)$$

$$E\mathbf{b}_1 + \mathbf{p}a_1 + i\mathbf{p} \times \mathbf{b}_1 = m\mathbf{b}_2. \quad (8.48)$$

Every operator acts on the following coefficient. Each entry still has the two independent complex components of (8.6). Appendix A.4 converts the system to the conventional four-component Dirac equation [6, pp. 101, 137–139].

8.4.4 Interacting factorization and reconstruction

Recall the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. For any complex paravector field $\psi(t, \mathbf{r})$, the product rule gives

$$\partial^*(\Phi^*\psi) + \Phi\partial\psi = (S + F)\psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi, \quad (8.49)$$

$$\partial(\Phi\psi) + \Phi^*\partial^*\psi = (S + F^*)\psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi. \quad (8.50)$$

The difference of (8.49) and (8.50) is $(F - F^*)\psi = 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi$.

Applying the block map $\mathbf{W}$ twice gives

$$\begin{aligned} \mathbf{W}(\partial)\mathbf{W}(\Phi^*)\Psi &= \begin{Bmatrix} \partial(\Phi\Psi_1) \\ \partial^*(\Phi^*\Psi_2) \end{Bmatrix}, \\ \mathbf{W}(\Phi^*)\mathbf{W}(\partial)\Psi &= \begin{Bmatrix} \Phi^*\partial^*\Psi_1 \\ \Phi\partial\Psi_2 \end{Bmatrix}, \\ \mathbf{W}(\Phi^*)^2 &= \det \Phi \mathbf{I}. \end{aligned}$$

Differentiating a parenthesized product includes the potential derivative. Define the interacting mass-conjugate blocks

$$\begin{aligned} \mathbf{D} &:= \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}, \\ \mathbf{D}' &:= \mathbf{W}(i\partial - q\Phi) + m\mathbf{I}. \end{aligned}$$

Multiplying and using the preceding mixed-derivative identities gives

$$\mathbf{D}'\mathbf{D}\Psi = - \begin{Bmatrix} (-E^2 + \mathbf{p}^2 + m^2 + iqF^*)\Psi_1 \\ (-E^2 + \mathbf{p}^2 + m^2 + iqF)\Psi_2 \end{Bmatrix}. \quad (8.51)$$

This recovers (8.35) at $q = 0$. With coupling, its entries obey

$$\begin{aligned} ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2) \Psi_1 &= 0, \\ ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2) \Psi_2 &= 0. \end{aligned}$$

Expanding gives the exact second-order equations

$$(-E^2 + \mathbf{p}^2 + m^2 + iqF^*)\Psi_1 = 0, \quad (8.52)$$

$$(-E^2 + \mathbf{p}^2 + m^2 + iqF)\Psi_2 = 0. \quad (8.53)$$

For the spin-term components, write $\mathbf{C} = \mathbf{E} + i\mathbf{B}$ and $\psi = a + \mathbf{b} \cdot \boldsymbol{\sigma}$, with complex functions $a : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\mathbf{b}, \mathbf{C} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Here $\mathbf{C}^*$ conjugates its scalar coefficients. Sigma multiplication gives

$$iqF\psi = iq\mathbf{C} \cdot \mathbf{b} + (iq\mathbf{C} - q\mathbf{C} \times \mathbf{b}) \cdot \boldsymbol{\sigma}, \quad (8.54)$$

$$iqF^*\psi = -iq\mathbf{C}^* \cdot \mathbf{b} + (-iq\mathbf{C}^* + q\mathbf{C}^* \times \mathbf{b}) \cdot \boldsymbol{\sigma}. \quad (8.55)$$

To construct solutions, take a complex paravector amplitude ψ of (8.6) satisfying $(-E^2 + \mathbf{p}^2 + m^2 + iqF)\psi = 0$. Set

$$\boxed{\Psi = \begin{Bmatrix} (E - \mathbf{p} \cdot \boldsymbol{\sigma})\psi/m \\ \psi \end{Bmatrix}}. \quad (8.56)$$

The first Dirac equation holds by construction; the second is the seed equation. Every massive solution has this form with $\psi = \Psi_2$. One must solve that seed equation for the chosen potential: a scalar KG seed lacks the spin term, and independently solved squared entries need not satisfy the first-order coupling.

8.4.5 Gauge phase

Recall the real scalar $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$ and gauge transformation $\Phi' = \Phi + \partial^*\lambda$, or $V' = V + \partial_t\lambda$, $\mathbf{A}' = \mathbf{A} - \partial_r\lambda$. The extra derivatives of λ are canceled by differentiating a scalar phase. Write $\Psi = e^{iq\lambda}\Psi'$; entrywise,

$$\begin{aligned} (i\partial - q\Phi^*)(e^{iq\lambda}\Psi'_k) &= e^{iq\lambda}(i\partial - q\partial\lambda - q\Phi^*)\Psi'_k \\ &= e^{iq\lambda}(i\partial - q\Phi'^*)\Psi'_k. \end{aligned}$$

The same product rule for $i\partial^* - q\Phi$ gives the other block entry. Thus the new amplitude is

$$\Psi' = e^{-iq\lambda}\Psi. \quad (8.57)$$

With $E' = i\partial_t - qV'$ and $\mathbf{p}' = -i\partial_r - q\mathbf{A}'$, the same product rule gives

$$\begin{aligned} E'\Psi' &= e^{-iq\lambda}E\Psi, \\ \mathbf{p}'\Psi' &= e^{-iq\lambda}\mathbf{p}\Psi. \end{aligned}$$

Consequently

$$\begin{aligned} &(\mathbf{W}(i\partial - q\Phi'^*) - m\mathbf{I})\Psi' \\ &= e^{-iq\lambda}(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi. \end{aligned} \quad (8.58)$$

Equation (8.58) preserves the Dirac equation; the phase sign is fixed by the potential shift [6, p. 137]. Its modulus is one, so the norm is unchanged. The same product rules preserve the KG and Pauli equations.

9 Low-energy limits

Recall the real mass $m > 0$, charge q , and potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The mechanical operators are $E = i\partial_t - qV$ and $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$; the fields are $\mathbf{E} = -\partial_{\mathbf{r}}V - \partial_t\mathbf{A}$ and $\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}$. Remove the rest oscillation, then iterate the residual energy. This gives Pauli from Dirac and Schrödinger from KG at leading order.

9.1 Exact reduction and the energy recursion

The Dirac entries Ψ_1, Ψ_2 are complex paravector amplitudes of (8.6), depending on real $(t, \mathbf{r})$. At free positive-energy rest, the time equation is

$$i\partial_t \Psi_k = m \Psi_k \implies \Psi_k(t) = e^{-imt} \Psi_k(0).$$

A free mode of real energy ω near m therefore factors as

$$e^{-i\omega t} = e^{-imt} e^{-i(\omega-m)t}.$$

The envelope has the smaller frequency $\omega - m$. Write $\Psi_k = e^{-imt} \Psi'_k$, or equivalently

$$\Psi'_k := e^{imt} \Psi_k, \quad k = 1, 2. \quad (9.1)$$

The positive sign removes the rest oscillation by its inverse. The product rule gives

$$(i\partial_t - qV)(e^{-imt} \Psi'_k) = e^{-imt} (m + i\partial_t - qV) \Psi'_k.$$

Hence

$$\begin{aligned} E \Psi_k &= e^{-imt} (E + m) \Psi'_k, \\ \mathbf{p} \Psi_k &= e^{-imt} \mathbf{p} \Psi'_k. \end{aligned}$$

The potentials are unchanged; E on the envelope measures mechanical energy above rest. Phase removal is exact and preserves the norm and both branches. The approximation below selects slow positive-energy envelopes: a free mode near $-m$ instead has residual frequency near $-2m$.

The coupled equations (8.42) and (8.43), after division by e^{-imt} , become

$$\begin{aligned} (E + m - \mathbf{p} \cdot \boldsymbol{\sigma}) \Psi'_2 &= m \Psi'_1, \\ (E + m + \mathbf{p} \cdot \boldsymbol{\sigma}) \Psi'_1 &= m \Psi'_2. \end{aligned}$$

Eliminate Ψ'_1 before expanding the ordered product:

$$((E + m + \mathbf{p} \cdot \boldsymbol{\sigma})(E + m - \mathbf{p} \cdot \boldsymbol{\sigma}) - m^2) \Psi'_2 = 0.$$

Using (8.16) and $[\mathbf{p} \cdot \boldsymbol{\sigma}, E] = -iq\mathbf{E} \cdot \boldsymbol{\sigma}$ gives

$$(E^2 + 2mE - \mathbf{p}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma}) \Psi'_2 = 0. \quad (9.2)$$

Rearranging gives the residual-energy recursion, with the operator order retained:

$$E \Psi'_2 = \frac{\mathbf{p}^2 - E^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} - [\mathbf{p} \cdot \boldsymbol{\sigma}, E]}{2m} \Psi'_2. \quad (9.3)$$

The imaginary electric term in (9.3) belongs to one chiral entry, whose norm alone is not the Dirac probability. To derive a probability-preserving Pauli equation, keep both entries.

At free positive-energy rest, $\Psi'_1 = \Psi'_2$: their sum survives and their difference vanishes. Define the corresponding complex paravector amplitudes

$$\phi := \frac{\Psi'_1 + \Psi'_2}{\sqrt{2}}, \quad (9.4)$$

$$\chi := \frac{\Psi'_2 - \Psi'_1}{\sqrt{2}}. \quad (9.5)$$

Adding and subtracting the coupled equations gives the exact pair

$$E\phi = (\mathbf{p} \cdot \boldsymbol{\sigma})\chi, \quad (9.6)$$

$$(E + 2m)\chi = (\mathbf{p} \cdot \boldsymbol{\sigma})\phi. \quad (9.7)$$

Their conserved norm is $\int (\|\phi\|^2 + \|\chi\|^2) d^3\mathbf{r}$. Rearrange the second equation for iteration:

$$\chi = \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})\phi}{2m} - \frac{E\chi}{2m}. \quad (9.8)$$

Start with the first term and substitute repeatedly into (9.8), replacing $E\phi$ with (9.6). Keep the operator order. For a real small parameter $0 < v \ll 1$, assume $\mathbf{p}/m = O(v)$, $E/m = O(v^2)$, $qV/m = O(v^2)$, $q\mathbf{B}/m^2 = O(v^2)$, and $q\mathbf{E}/m^2 = O(v^3)$, with smooth fields varying on scales $1/(mv)$ in space and $1/(mv^2)$ in time or longer. Then $\chi = O(v\phi)$ and each iteration adds a relative $O(v^2)$.

9.2 Leading order: Pauli

Recall ϕ, χ from (9.4) and (9.5). Keeping the first iterate gives

$$\begin{aligned} \chi &= \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{2m} \phi + O(v^3 \phi), \\ E\phi &= \frac{(\mathbf{p} \cdot \boldsymbol{\sigma})^2}{2m} \phi + O(mv^4 \phi). \end{aligned}$$

The product identity $(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}$ from (8.16) gives Pauli's equation at this order [7, pp. 35–36]:

$$\boxed{i\partial_t \phi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi.} \quad (9.9)$$

Here $\int \text{sc}(\phi^H \phi) d^3\mathbf{r} = 1$ at leading order. Its two complex components are the Pauli amplitudes; see Appendix A.4. It is the ψ used in the spin discussion. Equation (10.2) identifies spin as $\boldsymbol{\sigma}/2$, so the magnetic energy is $-(q/m)\mathbf{B} \cdot (\boldsymbol{\sigma}/2)$: magnetic moment equals q/m times spin, so the gyromagnetic factor is 2 before radiative corrections.

9.2.1 Components along a general field direction

For a nonzero static uniform field, let $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$ be its [real unit direction](#) and $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ its spin projectors. The complex paravector fields $\phi_{\pm} = \Pi_{\pm} \phi$ obey

$$i\partial_t \phi_{\pm} = \left(\frac{\mathbf{p}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi_{\pm}. \quad (9.10)$$

For a varying field direction, the projectors depend on $(t, \mathbf{r})$. Let H denote the Pauli Hamiltonian in (9.9). The product rule gives

$$\begin{aligned} i\partial_t(\Pi_\pm\phi) &= i(\partial_t\Pi_\pm)\phi + \Pi_\pm H\phi \\ &= H(\Pi_\pm\phi) + \left(i\partial_t\Pi_\pm + \frac{[\Pi_\pm, \mathbf{p}^2]}{2m}\right)\phi. \end{aligned} \quad (9.11)$$

Only the projector derivatives and kinetic commutator couple the sectors; the magnetic term commutes with its own projectors.

9.3 The same leading order: Schrödinger

When $\mathbf{B} = 0$, each complex coefficient of the Pauli amplitude obeys the scalar Schrödinger equation. KG gives the same result directly. Keep the operators $E, \mathbf{p}$ defined above. For the full scalar KG field $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$, remove the rest oscillation as in (9.1), defining its scalar envelope $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ by

$$\psi'(t, \mathbf{r}) = e^{-imt}\psi(t, \mathbf{r}). \quad (9.12)$$

The product rule gives $E\psi' = e^{-imt}(E + m)\psi$. Since m is constant, applying E again gives $E^2\psi' = e^{-imt}(E + m)^2\psi$; spatial derivatives leave the phase untouched. Substitution into (8.24) therefore gives

$$\begin{aligned} 0 &= (-E^2 + \mathbf{p}^2 + m^2)\psi' \\ &= e^{-imt}(-(E + m)^2 + \mathbf{p}^2 + m^2)\psi \\ &= e^{-imt}(-E^2 - 2mE + \mathbf{p}^2)\psi. \end{aligned}$$

The rest terms cancel. Divide by the nonzero phase to obtain

$$E\psi = \frac{\mathbf{p}^2 - E^2}{2m}\psi. \quad (9.13)$$

On a slow positive-energy envelope, $E/m = O(v^2)$, so $E^2\psi$ is smaller than $2mE\psi$ by $O(v^2)$. Dropping it at leading order gives [6, pp. 41–42, 139]

$$\boxed{i\partial_t\psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV\right)\psi.} \quad (9.14)$$

With $\mathbf{A} = 0$, its kinetic operator is $-\partial_{\mathbf{r}}^2/(2m)$. Schrödinger and Pauli therefore occur at the same leading order $1/m$; retaining spin gives the magnetic term in (9.9).

9.4 The next free order

For the free complex scalar envelope $\psi(t, \mathbf{r})$, set $qV = 0$ and $q\mathbf{A} = 0$, so E and $\mathbf{p}$ commute. On the positive-frequency KG branch, solving (9.13) gives

$$E\psi = \left(\sqrt{m^2 + \mathbf{p}^2} - m\right)\psi. \quad (9.15)$$

The same energy acts on either normalized positive-energy Dirac spin component. Iteration starts with $E\psi = \mathbf{p}^2\psi/(2m)$; its square supplies the next term:

$$E\psi = \left(\frac{\mathbf{p}^2}{2m} - \frac{\mathbf{p}^4}{8m^3} + O\left(\frac{\mathbf{p}^6}{m^5}\right)\right)\psi. \quad (9.16)$$

10 Spin and probability

Spin specifies the intrinsic rotation of an amplitude, comparing corresponding rotated points of real spacetime. A scalar KG amplitude stays unchanged and has spin zero; a two-component amplitude has the one-sided law below. The wave equation alone does not fix spin: free Dirac components also satisfy KG.

10.1 Spin one-half and rotations

Recall the complex paravector amplitude $\psi(t, \mathbf{r})$ of (8.6). For a **real unit axis** $\mathbf{u}$ and real angle θ , use the unitary rotor $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ of (3.55). The spinor law is

$$\psi' = R^H \psi. \quad (10.1)$$

Writing its infinitesimal action as $\psi' = \psi - i\theta S(\mathbf{u})\psi + O(\theta^2)$ defines the spin operator along $\mathbf{u}$:

$$S(\mathbf{u})\psi = \frac{1}{2}(\mathbf{u} \cdot \boldsymbol{\sigma})\psi. \quad (10.2)$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, its two spin projections are $\pm 1/2$ in units $\hbar = 1$. A 2π rotation gives $R = -1$, reversing the amplitude's sign; a 4π rotation restores it. Probabilities are unchanged by the overall sign, while a relative sign can be detected by interference [8, Sec. 6-3]. Each Dirac entry obeys this same rotation law.

For a real paravector observable Z , set $Z' = R^H Z R$. Recall $\rho = \|\psi\|^2 > 0$ and the scalar expectation (8.4). Since $RR^H = 1$,

$$\langle Z' \rangle_{\psi'} = \frac{\text{sc}(\psi^H R R^H Z R R^H \psi)}{\rho} = \langle Z \rangle_{\psi}. \quad (10.3)$$

Thus rotating both amplitude and observable preserves the measurement; $Z = \mathbf{u}' \cdot \boldsymbol{\sigma}$ recovers the vector rotation of (4.11) for any real unit vector $\mathbf{u}'$.

10.2 Projectors and probabilities

For a real unit measurement axis $\mathbf{u}$, recall the real projectors

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (10.4)$$

Their algebra in Section 3.7 gives

$$\begin{aligned} \psi &= \Pi_+ \psi + \Pi_- \psi, \\ S(\mathbf{u})(\Pi_{\pm} \psi) &= \pm \frac{1}{2} \Pi_{\pm} \psi, \\ \langle \Pi_+ \psi \mid \Pi_- \psi \rangle &= 0, \\ \|\Pi_{\pm} \psi\|^2 &= \text{sc}(\psi^H \Pi_{\pm} \psi). \end{aligned}$$

The Born probability is the projected squared norm divided by $\rho = \|\psi\|^2$. In the general-axis orthonormal basis (8.5), $\psi = c_+ \psi_+ + c_- \psi_-$ with complex scalars $c_{\pm}$, so $\rho = \|c_+\|^2 + \|c_-\|^2$ and the probabilities are $\|c_{\pm}\|^2/\rho$. The same-side and cross-side projector products are already derived in Section 3.7; the ladder paravectors in (3.44) exchange the two sectors.

To evaluate the probabilities directly from paravector components, define

$$B := \frac{\psi\psi^H}{\rho} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}. \quad (10.5)$$

Here B is real and $\mathbf{s} \in \mathbb{R}^3$ is the polarization. The basis construction gives $\det \psi = 0$ (Appendix A.4), hence $0 = \det B = 1 - \mathbf{s}^2$: the polarization is a real unit vector. Cyclicity of the scalar part gives

$$\|\Pi_{\pm}\psi\|^2 = \text{sc}(\psi^H \Pi_{\pm} \psi) = \rho \text{sc}(\Pi_{\pm} B) = \frac{1}{2}\rho(1 \pm \mathbf{u} \cdot \mathbf{s}).$$

Therefore

$$p_{\pm}(\mathbf{u}) = \frac{\|\Pi_{\pm}\psi\|^2}{\rho} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (10.6)$$

The mean spin along $\mathbf{u}$ is $(p_+ - p_-)/2 = \mathbf{u} \cdot \mathbf{s}/2$. For a spatial wavefunction, (10.6) is conditional on position; integrate numerator and denominator to obtain the whole-packet probability. If the angle between $\mathbf{u}$ and $\mathbf{s}$ is the real angle θ ,

$$p_+ = \cos^2(\theta/2), \quad p_- = \sin^2(\theta/2).$$

For $R = \exp(i\theta \mathbf{a} \cdot \boldsymbol{\sigma}/2)$, with real unit rotation axis $\mathbf{a}$ and real angle θ , the amplitude law (10.1) gives

$$B' = R^H B R. \quad (10.7)$$

Thus the polarization rotates as a vector although its amplitude has spin one-half. Rodrigues' formula gives the original-axis overlap:

$$\mathbf{s} \cdot \mathbf{s}' = (\mathbf{a} \cdot \mathbf{s})^2 + (1 - (\mathbf{a} \cdot \mathbf{s})^2) \cos \theta.$$

For $\mathbf{a} \cdot \mathbf{s} = 0$, the probability to retain the initial spin is $\cos^2(\theta/2)$.

10.3 Spin uncertainty

Recall the real polarization $\mathbf{s}$ in (10.5). For real unit axes $\mathbf{u}, \mathbf{u}'$, set $A = \mathbf{u} \cdot \boldsymbol{\sigma}/2$ and $A' = \mathbf{u}' \cdot \boldsymbol{\sigma}/2$. Define the real variance by $\text{var}(A) = \langle A^2 \rangle - \langle A \rangle^2$. The sigma product gives

$$\begin{aligned} [A, A'] &= \frac{i}{2}(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \text{var}(A) &= \frac{1 - (\mathbf{u} \cdot \mathbf{s})^2}{4}. \end{aligned}$$

For a normalized amplitude, positivity of $\|(A - \langle A \rangle)\psi + i\lambda(A' - \langle A' \rangle)\psi\|^2$ for every real scalar λ makes its quadratic discriminant nonpositive. This yields the Robertson bound

$$\text{var}(A) \text{var}(A') \geq \frac{1}{4} \|\langle [A, A'] \rangle\|^2 = \frac{((\mathbf{u} \times \mathbf{u}') \cdot \mathbf{s})^2}{16}. \quad (10.8)$$

11 Rotations and Lorentz boosts

We now apply the algebraic actions of Section 4 to the physical variables already defined. Primed fields are evaluated at X' and unprimed fields at X . Factors are constant until the final subsection.

11.1 Covariance of the physical equations

Let T be a complex [unit paravector](#) with $\det T = 1$. It may be any ordered product of $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with [real unit vector](#) $\mathbf{u}$ and real angle or rapidity θ . Recall the real event X , momentum P , potential Φ , and electric current D . Each transforms by

$$Z' = T^{-1} Z (T^{-1})^H, \quad Z = X, P, \Phi, D. \quad (11.1)$$

The derivative and complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ use the corresponding laws

$$\partial' = T^H \partial T, \quad (11.2)$$

$$F' = T^{-1} F T. \quad (11.3)$$

Equation (11.1) acts on real four-vectors; (11.2) is the chain rule. For a differentiable scalar $f : \mathbb{R}^4 \rightarrow \mathbb{C}$ with $f'(X') = f(X)$, cyclicity gives

$$df = \text{sc}((\partial f) dX) = \text{sc}((T^H(\partial f)T) dX').$$

Since this holds for every real increment dX' , the primed derivative is $\partial' f' = T^H(\partial f)T$ [3, p. 101].

The potential identity (7.11), with real scalar $S = \text{sc}(\partial\Phi)$, then gives

$$\partial' \Phi' = T^H(S + F^*)(T^H)^{-1}.$$

The scalar part remains S ; sigma conjugation of the vector part yields the field similarity (11.3). Since $T^* = (T^H)^{-1}$, Maxwell covariance follows by cancellation:

$$\partial' F' = T^H(\partial F)T = T^H D^* T = D'^*. \quad (11.4)$$

This recovers (7.2) in the primed frame. Adjagation of P' and scalar-part cyclicity give

$$\text{sc}(\text{adj}(P') X') = \text{sc}(T^H \text{adj}(P) X (T^H)^{-1}) = \text{sc}(\text{adj}(P) X). \quad (11.5)$$

Thus the plane wave (8.9) satisfies $f(X', P') = f(X, P)$.

11.2 Moving frames, clock rates, and lengths

Recall $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , [real unit vector](#) $\mathbf{u}$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. Here γ is the factor for frame velocity $\beta \mathbf{u}$. The explicit gamma form is (3.71). For real $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, take coincident inertial origins at $t = t' = 0$. The passive boost $X' = L^{-1} X L^{-1}$ means that the primed frame moves at $+\beta \mathbf{u}$. Equations (4.13)–(4.14) become

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (11.6)$$

$$\mathbf{r}' = \mathbf{r}_\perp + \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad (11.7)$$

where $\mathbf{r}_\parallel = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ and $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$ [3, p. 96, eqs. (5.1)–(5.2)]. Reversing θ gives $X = L X' L$.

11.2.1 Spacetime increments and velocity

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$, along a real world line $X(s)$. For constant L , differentiation gives

$$dX' = L^{-1} dX L^{-1}. \quad (11.8)$$

Because ds is invariant, $U' = L^{-1}UL^{-1}$. At a point where $U = L^2$, this gives $U' = 1$ and $dX' = ds$: the boost takes the particle's instantaneous tangent to rest. For varying velocity, a different L is needed at each point; Section 11.4 supplies its derivative terms.

Recall the particle velocity $\mathbf{v} = d\mathbf{r}/dt$ and resolve it along $\mathbf{u}$. Dividing the spatial increment by $dt' = \gamma(1 - \beta\mathbf{u} \cdot \mathbf{v}) dt$ gives

$$\mathbf{v}' = \frac{\mathbf{v}_\perp/\gamma + \mathbf{v}_\parallel - \beta\mathbf{u}}{1 - \beta\mathbf{u} \cdot \mathbf{v}}. \quad (11.9)$$

Interval invariance also gives

$$1 - \mathbf{v}'^2 = \frac{1 - \mathbf{v}^2}{\gamma^2(1 - \beta\mathbf{u} \cdot \mathbf{v})^2}. \quad (11.10)$$

Thus (11.9) preserves subluminal motion and light speed [2, pp. 135–136, eqs. (7.22)–(7.23)]. The constant-factor law (11.8) also commutes with the world-line integral (6.14).

11.2.2 Clock rates and measured lengths

For a clock at rest in the primed frame, (11.6) uses $d\mathbf{r} = \beta\mathbf{u} dt$, hence

$$dt' = \gamma(1 - \beta^2) dt = dt/\gamma. \quad (11.11)$$

Here $dt' = ds$; finite elapsed time follows from (6.15) [2, pp. 118–120]. For a rod at rest in the unprimed frame, let $\Delta\mathbf{r}$ be its endpoint separation. A simultaneous measurement in the primed frame requires $\Delta t' = 0$, so $\Delta t = \beta\mathbf{u} \cdot \Delta\mathbf{r}$. Substitution gives

$$\Delta\mathbf{r}'_\parallel = \Delta\mathbf{r}_\parallel/\gamma. \quad (11.12)$$

The transverse separation remains unchanged by (11.7). Thus longitudinal contraction is measured at equal primed times [2, pp. 122–123]. Conversely, equal unprimed times give $\Delta t' = -\gamma\beta\mathbf{u} \cdot \Delta\mathbf{r}$; simultaneity is frame-dependent.

11.2.3 Composing frames

Collinear boosts add rapidities by (3.75). For a second real rapidity θ' , set $\beta' = \tanh\theta'$ and $\gamma' = \cosh\theta'$. Their combined parameters are

$$\beta'' = \tanh(\theta + \theta') = \frac{\beta + \beta'}{1 + \beta\beta'}, \quad (11.13)$$

$$\gamma'' = \gamma\gamma'(1 + \beta\beta'). \quad (11.14)$$

These are the collinear velocity and clock-factor composition laws. Noncollinear boosts instead have the rotation commutator (3.76), producing the Thomas–Wigner rotation.

11.2.4 Energy and momentum

For the real momentum $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, the same congruence gives

$$E' = \gamma(E - \beta \mathbf{u} \cdot \mathbf{p}), \quad (11.15)$$

$$\mathbf{p}' = \mathbf{p}_\perp + \gamma(\mathbf{p}_\parallel - \beta E \mathbf{u}). \quad (11.16)$$

Here the projections are along the boost axis $\mathbf{u}$. Both coefficients change, but $E'^2 - \mathbf{p}'^2 = m^2$ leaves rest mass unchanged [2, pp. 136–138]. The energy increase and force response are derived in (6.12)–(6.13).

11.3 Boosted fields and energy flux

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with the parameters above, both the field coefficients and their spacetime arguments transform:

$$F'(X') = L^{-1}F(LX'L)L. \quad (11.17)$$

Thus $F'(X') = L^{-1}F(X)L$ at corresponding events $X = LX'L$. Apply (4.15) to $\mathbf{E} + i\mathbf{B}$ and separate its real and imaginary parts:

$$\mathbf{E}' = \mathbf{E}_\parallel + \gamma(\mathbf{E}_\perp + \beta \mathbf{u} \times \mathbf{B}), \quad (11.18)$$

$$\mathbf{B}' = \mathbf{B}_\parallel + \gamma(\mathbf{B}_\perp - \beta \mathbf{u} \times \mathbf{E}). \quad (11.19)$$

Parallel components are unchanged; the projections use the same [real unit axis](#) as (11.7) [3, p. 108]. Recall the stress map $T(N) = FN F^H/2$, with real paravector input N . Since $N' = L^{-1}NL^{-1}$,

$$T'(N') = \frac{1}{2}F'N'F'^H = L^{-1}T(N)L^{-1}. \quad (11.20)$$

Here T' denotes the map formed from F' . Each energy–momentum flux has the real four-vector law; $T(1)$ alone is only the flux through the chosen time direction. The same cancellation in (7.26) gives the transformed proper force.

11.4 Varying frames and local bases

Let $T(s)$ be a smooth complex [unit paravector](#) of [determinant one](#) along a real parameter s . When s is proper time, recall the real tangent $U = dX/ds$, a unit paravector with $UU^{-1} = 1$. Define the complex paravector $\Omega(s) := T^{-1}dT/ds$. The inverse rule gives

$$\frac{dT^{-1}}{ds} = -\Omega T^{-1}.$$

For a complex paravector $Z(s)$ transformed by similarity, $Z' = T^{-1}ZT$, product differentiation yields

$$\frac{dZ'}{ds} = T^{-1} \frac{dZ}{ds} T + [Z', \Omega]. \quad (11.21)$$

Differentiating again and replacing $T^{-1}(dZ/ds)T$ by its first-derivative expression gives

$$\begin{aligned} \frac{d^2 Z'}{ds^2} &= T^{-1} \frac{d^2 Z}{ds^2} T + 2 \left[\frac{dZ'}{ds}, \Omega \right] \\ &\quad + \left[Z', \frac{d\Omega}{ds} \right] + [\Omega, [Z', \Omega]]. \end{aligned} \quad (11.22)$$

The last term follows from $-[[Z', \Omega], \Omega] = [\Omega, [Z', \Omega]]$. For a real paravector transformed instead by congruence, $Z' = T^{-1}Z(T^{-1})^H$, the same rule gives

$$\frac{dZ'}{ds} = T^{-1} \frac{dZ}{ds} (T^{-1})^H - \Omega Z' - Z' \Omega^H. \quad (11.23)$$

Equations (11.21) and (11.23) include the frame terms omitted by a constant-factor transformation. For a varying rotation $T = R = \exp(i\theta(s)\mathbf{u}(s) \cdot \boldsymbol{\sigma}/2)$, with real $\theta(s)$ and **real unit vector** $\mathbf{u}(s)$, unitarity gives $\Omega^H = -\Omega$, so both formulas agree.

When s is proper time, applying the appropriate formula to the pointwise transformed tangent U gives its acceleration law. This differs from first transforming the event X and then differentiating: a varying event map already contributes frame terms to its first derivative. The similarity formula also gives the local algebra-basis law $\sigma'_k = T^{-1}\sigma_k T$, with

$$\frac{d\sigma'_k}{ds} = [\sigma'_k, \Omega], \quad k = 1, 2, 3. \quad (11.24)$$

This evolution preserves all generator products. A general complex local basis is nevertheless distinct from a real spacetime frame, whose vectors require the congruence law.

12 Flat spacetime in generalized coordinates

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be smooth functions on a coordinate chart with real coordinates $(t, \mathbf{r})$. They specify inertial coordinates through the real paravector

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

Along a future timelike path with proper time s , recall the real $U = dX/ds$, a **unit paravector** with $UU^{-1} = 1$. Require the Jacobian of $(f, \mathbf{g})$ to be nonsingular. Differentiating gives

$$\begin{aligned} dX &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}) dt + \sum_{k=1}^3 (\partial_{r_k} f + \partial_{r_k} \mathbf{g} \cdot \boldsymbol{\sigma}) dr_k, \\ ds^2 &= dX dX^* = df^2 - d\mathbf{g}^2. \end{aligned}$$

With $x_0 = t$, $x_k = r_k$ and $\partial_k = \partial/\partial x_k$, the interval is $ds^2 = \sum_{j,k=0}^3 g_{jk} dx_j dx_k$, with real coefficients

$$g_{jk} = \partial_j f \partial_k f - \partial_j \mathbf{g} \cdot \partial_k \mathbf{g}, \quad j, k = 0, 1, 2, 3. \quad (12.1)$$

This is the flat Minkowski metric in changed coordinates; varying coefficients do not imply curvature.

Along this future timelike curve, $df > 0$. Define its real inertial three-velocity $\mathbf{v} = d\mathbf{g}/df$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, with $\mathbf{v}^2 < 1$. Then

$$\begin{aligned} ds &= \sqrt{1 - \mathbf{v}^2} df, \\ U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Apply (6.7) with inertial time f and coordinate acceleration $d\mathbf{v}/df$. For inertial motion, the scalar part gives $\mathbf{v} \cdot d\mathbf{v}/df = 0$, and its vector part then gives $d\mathbf{v}/df = 0$. Therefore

$$\mathbf{g} = \mathbf{C}_1 f + \mathbf{C}_2. \quad (12.2)$$

The real constants $\mathbf{C}_1, \mathbf{C}_2 \in \mathbb{R}^3$ are velocity and initial position. If primes denote total derivatives along this curve with respect to t , then, wherever $f' \neq 0$,

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}''f' - \mathbf{g}'f''}{(f')^2},$$

$$[U, dU/ds] = 2i\gamma^2 \left(\mathbf{v} \times \frac{d\mathbf{v}}{ds} \right) \cdot \boldsymbol{\sigma}.$$

13 Magnetic duality and constant-field motion

This is the standard classical magnetic-source extension on real spacetime. Its magnetic charge is a hypothetical source parameter; no observed monopole is assumed. Setting it to zero recovers ordinary electrodynamics.

13.1 The imaginary force and magnetic charge

Recall the real proper velocity $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$, a [unit paravector](#) with $UU^{-1} = 1$, and real momentum $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, with constant mass $m > 0$. The real and imaginary parts of the product (7.25) are the electric force and its magnetic dual per unit charge. Let $q, q' \in \mathbb{R}$ be electric and magnetic charges; here the prime labels an independent charge. The classical dyon force is

$$\boxed{\frac{dP}{ds} = \text{re}((q - iq')FU)}. \quad (13.1)$$

Its real components are

$$\frac{dE}{dt} = (q\mathbf{E} + q'\mathbf{B}) \cdot \mathbf{v}, \quad (13.2)$$

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q'(\mathbf{B} - \mathbf{v} \times \mathbf{E}). \quad (13.3)$$

Equation (13.2) is the work rate; (13.3) contains the electric and magnetic forces. The minus sign in $q - iq'$ follows from $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$: a positive magnetic charge feels $+q'\mathbf{B}$ at rest. Both forces preserve the mass shell, since cyclicity and $UU^* = 1$ give

$$\text{sc}(U^*FU) = \text{sc}(FUU^*) = \text{sc}(F) = 0.$$

Its real and imaginary parts give the two tangency conditions separately.

Let $M = d + \mathbf{M} \cdot \boldsymbol{\sigma}$ be a real magnetic-current paravector, with real magnetic charge density $d(t, \mathbf{r})$ and current $\mathbf{M}(t, \mathbf{r}) \in \mathbb{R}^3$. Maxwell's extension is

$$\boxed{\partial F = D^* + iM^*}. \quad (13.4)$$

This retains the electric-source laws (7.3), (7.5), and (7.8). Its magnetic parts become

$$\partial_{\mathbf{r}} \cdot \mathbf{B} = d, \quad (13.5)$$

$$\partial_{\mathbf{r}} \times \mathbf{E} + \partial_t \mathbf{B} = -\mathbf{M}, \quad (13.6)$$

$$\partial_t d + \partial_{\mathbf{r}} \cdot \mathbf{M} = 0. \quad (13.7)$$

The divergence and curl in (13.5) and (13.6) imply magnetic-charge conservation (13.7). For any real constant angle a , the classical equations admit the simultaneous duality transformation

$$F \mapsto e^{-ia} F, \quad (13.8)$$

$$q + iq' \mapsto e^{-ia}(q + iq'), \quad (13.9)$$

$$D + iM \mapsto e^{-ia}(D + iM). \quad (13.10)$$

Equations (13.8) and (13.10) multiply both sides of Maxwell's law by the same phase; (13.9) leaves $(q - iq')F$ unchanged. A specified quantum charge spectrum need not admit every angle. Since a smooth curl has zero divergence, magnetic charge requires patched, singular, or enlarged potentials [9, pp. 6–8]. This extension uses real spacetime.

13.2 Exact motion in a constant field

Recall the real proper velocity $U = dX/ds$, a unit paravector with $UU^{-1} = 1$, real charges q, q' , mass $m > 0$, and momentum $P = mU$. For constant complex electromagnetic field F , define the complex unit paravector

$$T(s) = \exp\left(\frac{(q - iq')s}{2m} F\right). \quad (13.11)$$

The solution for future real initial momentum $P(0)$ with $\det P(0) = m^2$ is

$$P(s) = T(s)P(0)T(s)^H. \quad (13.12)$$

Differentiating, with real $U = P/m$, gives

$$\begin{aligned} \frac{dP}{ds} &= \frac{(q - iq')FP + P((q - iq')F)^H}{2m} \\ &= \operatorname{re}((q - iq')FU). \end{aligned}$$

This congruence preserves Hermiticity, determinant, and the future cone; $\det T = 1$ follows from $\operatorname{sc}(F) = 0$. Hence $U = P/m$ remains a physical proper velocity. For $q' = 0$, real unit vector $\mathbf{u}$, and constant real field magnitudes E, B ,

$$\begin{aligned} T(s) &= \exp(iqBs\mathbf{u} \cdot \boldsymbol{\sigma}/(2m)) & (\mathbf{E} = 0, \mathbf{B} = B\mathbf{u}), \\ T(s) &= \exp(qEs\mathbf{u} \cdot \boldsymbol{\sigma}/(2m)) & (\mathbf{B} = 0, \mathbf{E} = E\mathbf{u}). \end{aligned}$$

These are $R = \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with $\theta = qBs/m$ and $L = \exp(\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with $\theta = qEs/m$. The evolution changes momentum in a fixed frame. This explains the reversed sandwiches relative to the coordinate actions (4.10) and (4.12). For the electric case with $qE > 0$, $X(0) = 0$, and $P(0) = m$, one has $U = T^2$. Integrating gives (6.17) with $a = qE/m$ [3, p. 117]. Complex field factors thus already generate real motion.

14 Extrapolation and speculation

The following coordinate and force laws are additional hypotheses. Lorentz covariance, the ordinary force law, stability, and the cosmological stress tensor provide established tests of these toy models. The tests do not identify a dark sector or an electroweak interaction with imaginary coordinates.

14.1 Testing a complex momentum equation

Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. Keep real charges q, q' and complex electromagnetic field F as above, but postulate a complex paravector $P(s)$ with

$$\frac{dP}{ds} = (q + iq')FU. \quad (14.1)$$

Extracting real and imaginary parts gives

$$\begin{aligned} \frac{d \operatorname{re}(P)}{ds} &= q \operatorname{re}(FU) - q' \operatorname{im}(FU), \\ \frac{d \operatorname{im}(P)}{ds} &= q \operatorname{im}(FU) + q' \operatorname{re}(FU). \end{aligned}$$

The real magnetic force has the opposite sign to (13.1). Even when $q' = 0$, the imaginary force $q \operatorname{im}(FU)$ remains. At rest in a pure magnetic field, $FU = i\mathbf{B} \cdot \boldsymbol{\sigma}$: the proposed law produces imaginary momentum where an electric charge stays at rest. Thus the identification $P = mU$ with real U is inconsistent in general. If P is independent of the real trajectory, these equations only accumulate two impulses; they do not prescribe a complex trajectory.

To close a complex model, replace the real tangent U by P/m , with complex P and real $m > 0$. Now $s \in \mathbb{R}$ is an evolution parameter, not established physical proper time. The law becomes

$$\frac{dP}{ds} = \frac{q + iq'}{m} FP.$$

It preserves the complex determinant because

$$\begin{aligned} \frac{d \det P}{ds} &= 2 \operatorname{sc}(\operatorname{adj}(P)) \frac{dP}{ds} \\ &= \frac{2(q + iq')}{m} \det P \operatorname{sc}(F) = 0. \end{aligned}$$

For constant F , the complete solution is

$$P(s) = \exp\left(\frac{(q + iq')s}{m} F\right) P(0). \quad (14.2)$$

Take $q' = 0$, $q \neq 0$, $\mathbf{E} = 0$, and constant $\mathbf{B} = B\mathbf{u}$ with real $B \neq 0$ and arbitrary real unit $\mathbf{u}$. With $P(0) = m$, this gives

$$P(s) = m \cos(qBs/m) + im \sin(qBs/m) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (14.3)$$

Integrating for a complex coordinate $Z(s)$ defined by $dZ/ds = P/m$ and $Z(0) = 0$ gives

$$Z(s) = \frac{m}{qB} \sin(qBs/m) + i \frac{m}{qB} (1 - \cos(qBs/m)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (14.4)$$

The real energy in (14.3) changes sign, and the time in (14.4) ceases to increase, despite $\det P = m^2$. This fails the ordinary at-rest electric charge limit even with zero magnetic charge. Determinant conservation alone is insufficient for a physical mass shell or causality.

14.2 Testing complex spacetime and its null set

Let $Z = X + iY$ be a complex coordinate, with real paravectors $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ and $Y = u + \mathbf{w} \cdot \boldsymbol{\sigma}$. Here $t, u \in \mathbb{R}$ and $\mathbf{r}, \mathbf{w} \in \mathbb{R}^3$ are independent coordinates. Multiplication gives

$$\det Z = t^2 - \mathbf{r}^2 - u^2 + \mathbf{w}^2 + 2i(tu - \mathbf{r} \cdot \mathbf{w}). \quad (14.5)$$

Complex nullity requires both

$$\det X = \det Y, \quad tu = \mathbf{r} \cdot \mathbf{w}.$$

For arbitrary perpendicular real unit vectors $\mathbf{u}, \mathbf{a}$, $Z = (\mathbf{u} + i\mathbf{a}) \cdot \boldsymbol{\sigma}$ is complex-null but its real part is spacelike. Identifying every complex-null separation with observable light propagation would therefore permit real spacelike connections. Taking only the real determinant gives instead

$$\text{re}(\det(dZ)) = dt^2 - d\mathbf{r}^2 - du^2 + d\mathbf{w}^2. \quad (14.6)$$

This metric has signature (4, 4). Either proposal needs an operationally defined real causal slice. Complex field values on real spacetime require no such change of coordinates.

14.3 Separate masses and small couplings

Consider a real four-vector field $A(t, \mathbf{r}) = a(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$. Assign proposed real masses $m_0, m_1 \geq 0$ to its scalar and vector components:

$$\begin{aligned} (\square + m_0^2)a &= 0, \\ (\square + m_1^2)\mathbf{A} &= 0. \end{aligned}$$

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with arbitrary real unit $\mathbf{u}$, define $\beta = \tanh \theta$ and $\gamma = \cosh \theta$ for real rapidity θ . The passive boost gives

$$\begin{aligned} a' &= \gamma(a - \beta \mathbf{u} \cdot \mathbf{A}), \\ (\square' + m_0^2)a' &= -\gamma\beta(m_0^2 - m_1^2)\mathbf{u} \cdot \mathbf{A}. \end{aligned} \quad (14.7)$$

Covariance for arbitrary fields forces $m_0^2 = m_1^2$, and similarly for every spatial component. Distinct masses can label distinct fields, but cannot label time and space entries of one Lorentz vector.

As a separate toy model, mix two scalar fields. Let $z = a + ib$, with $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$, and let $m > 0$ and h be real constants; h has units of mass squared. The real Lagrangian density

$$L = \frac{1}{2} \det(\partial a) + \frac{1}{2} \det(\partial b) - \frac{1}{2} m^2 (a^2 + b^2) - hab \quad (14.8)$$

has positive time-derivative terms, since $\det(\partial a) = (\partial_t a)^2 - (\partial_{\mathbf{r}} a)^2$. The resulting equations and their diagonalizing real combinations are

$$\begin{aligned} (\square + m^2)a + hb &= 0, \\ (\square + m^2)b + ha &= 0, \\ a' &= (a + b)/\sqrt{2}, \\ b' &= (a - b)/\sqrt{2}. \end{aligned}$$

$$(\square + m^2 + h)a' = 0, \quad (14.9)$$

$$(\square + m^2 - h)b' = 0. \quad (14.10)$$

The squared masses are $m^2 \pm h$, both nonnegative when $|h| \leq m^2$. The mixing term breaks continuous $z \mapsto e^{ic}z$ symmetry ($c \in \mathbb{R}$ constant), but leaves the principal operator $\square$ and its light cone unchanged. Small coupling does not create complex coordinates. The electroweak interaction likewise requires additional internal gauge and chiral structure [10, pp. 180–182].

By contrast, the alternative kinetic term

$$\frac{1}{2} \operatorname{re}(\det(\partial z)) = \frac{1}{2} \det(\partial a) - \frac{1}{2} \det(\partial b)$$

has real kinetic Hamiltonian density H :

$$H = \frac{1}{2}((\partial_t a)^2 + (\partial_r a)^2) - \frac{1}{2}((\partial_t b)^2 + (\partial_r b)^2). \quad (14.11)$$

This density is unbounded below: the independent b mode is a ghost, whose negative kinetic sign persists at arbitrarily small coupling [11, Secs. III–V].

14.4 What would be needed for a dark sector?

A dark-matter model must supply abundance, lifetime, gravitational clustering, and interaction rates. Complex notation alone fixes none. To distinguish matter from accelerated expansion, additionally assume homogeneous general-relativistic cosmology. Let $\phi(t) \in \mathbb{R}$ be a canonical scalar field, $V(\phi)$ its real potential-energy density, $u(t)$ its energy density, and $p(t)$ its isotropic pressure. Its stress tensor gives [12, pp. 34–35]

$$u = \frac{1}{2}(\partial_t \phi)^2 + V, \quad (14.12)$$

$$p = \frac{1}{2}(\partial_t \phi)^2 - V. \quad (14.13)$$

For real mass $m > 0$, take $V = m^2 \phi^2/2$ and oscillations faster than cosmic expansion. Neglect expansion over one cycle and take $\phi = A \cos(mt)$ with real constant amplitude A . Cycle averaging gives

$$\begin{aligned} \langle \tfrac{1}{2}(\partial_t \phi)^2 \rangle &= \langle V \rangle = m^2 A^2/4, \\ \langle p \rangle &= 0. \end{aligned}$$

This limit behaves as pressureless matter. Neglecting kinetic energy instead gives $p = -u$. For the real positive cosmic scale factor $a(t)$ and Newton's constant $G > 0$, the acceleration equation is [12, p. 6]

$$\frac{\partial_t^2 a}{a} = -\frac{4\pi G}{3}(u + 3p). \quad (14.14)$$

Acceleration therefore requires $u + 3p < 0$, not merely an additional positive mass. A complex-spacetime hypothesis still needs a stable action, a physical causal structure, and distinct testable predictions.

A Pauli-matrix representation

This appendix translates the established paravector operations into matrices. Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Write $\mathbf{M} = \operatorname{mat}(Z)$, a 2×2 matrix with entries in $\mathbb{C}$. The identity matrices are $\mathbf{I}_2$ and $\mathbf{I}_4$ in their stated dimensions.

A.1 Faithful representation

The Pauli images of the generators are

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Consequently,

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix}. \quad (\text{A.1})$$

The generator images obey the sigma product rule and are linearly independent over $\mathbb{C}$ together with $\mathbf{I}_2$. The map is therefore faithful, preserves addition, sends 1 to $\mathbf{I}_2$, and satisfies

$$\text{mat}(Z'Z) = \text{mat}(Z')\text{mat}(Z). \quad (\text{A.2})$$

Here Z' is any second complex paravector. Thus algebra multiplication is ordinary matrix multiplication.

A.2 Conjugations and product order

For complex matrices $\mathbf{M}, \mathbf{N}$, transpose $^\top$ exchanges row and column indices; entrywise complex conjugation $^\mathbb{C}$ sends each $a + ib$ to $a - ib$ for real a, b . Their composition is $\mathbf{M}^\mathbf{H} = (\mathbf{M}^\mathbb{C})^\top$. Introduce the fixed real matrix

$$\mathbf{S} = \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}, \quad \mathbf{S}^{-1} = -\mathbf{S}.$$

The paravector sigma conjugation has the matrix image

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \mathbf{M}^\mathbb{C} \mathbf{S} =: \mathbf{M}^*. \quad (\text{A.3})$$

Substituting the basis in (A.3) gives

Algebra basis	1	i	σ_k	$i\sigma_k$
Hermitian image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$\text{mat}(\sigma_k)$	$-i \text{mat}(\sigma_k)$
Sigma-conjugate image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$-\text{mat}(\sigma_k)$	$i \text{mat}(\sigma_k)$

Here $k = 1, 2, 3$. These are exactly the operations of Section 2, so $\text{mat}(Z^\mathbf{H}) = \mathbf{M}^\mathbf{H}$. Their product-order rules are

$$\begin{aligned} (\mathbf{MN})^\top &= \mathbf{N}^\top \mathbf{M}^\top, \\ (\mathbf{MN})^\mathbb{C} &= \mathbf{M}^\mathbb{C} \mathbf{N}^\mathbb{C}, \\ (\mathbf{MN})^\mathbf{H} &= \mathbf{N}^\mathbf{H} \mathbf{M}^\mathbf{H}. \end{aligned}$$

For sigma conjugation the middle factors cancel:

$$(\mathbf{MN})^* = \mathbf{S}^{-1} \mathbf{M}^\mathbb{C} \mathbf{S} \mathbf{S}^{-1} \mathbf{N}^\mathbb{C} \mathbf{S}.$$

Thus sigma conjugation preserves product order:

$$(\mathbf{MN})^* = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.4})$$

This verifies (2.4) in matrices. Hermitian and sigma conjugation commute, and applying either twice restores the original matrix. The matrix-only operations can be recovered as

$$\begin{aligned} \mathbf{M}^{\mathbf{C}} &= \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1}, \\ \mathbf{M}^{\mathbf{T}} &= \mathbf{S} (\mathbf{M}^*)^{\mathbf{H}} \mathbf{S}^{-1}. \end{aligned}$$

A.3 Scalar part, determinant, and norm

For $\mathbf{M} = (m_{jk})$, its trace is $m_{11} + m_{22}$ and its determinant is $m_{11}m_{22} - m_{12}m_{21}$. Substituting (A.1) gives

$$\text{sc}(Z) = \frac{1}{2} \text{tr} \mathbf{M} = S, \quad (\text{A.5})$$

$$\begin{aligned} (\text{vec}(Z))_k &= \frac{1}{2} \text{tr}(\text{mat}(\sigma_k) \mathbf{M}) = V_k, \\ \det Z &= \det \mathbf{M} = S^2 - \mathbf{V}^2. \end{aligned} \quad (\text{A.6})$$

By (A.5), matrix-trace cyclicity is precisely the scalar-part cyclicity used in the main text. For $\mathbf{M}' = \text{mat}(Z')$,

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}' \mathbf{M}) = \text{sc}(ZZ').$$

The remaining extraction and norm formulas are

$$\begin{aligned} \text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) &= \mathbf{M} - \text{sc}(Z) \mathbf{I}_2, \\ \text{mat}(\text{re}(Z)) &= \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\mathbf{H}}), \end{aligned} \quad (\text{A.7})$$

$$\text{mat}(\text{im}(Z)) = (\mathbf{M} - \mathbf{M}^{\mathbf{H}})/(2i), \quad (\text{A.8})$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(\mathbf{M}^{\mathbf{H}} \mathbf{M}) = \|S\|^2 + \|\mathbf{V}\|^2. \quad (\text{A.9})$$

Equations (A.7)–(A.8) extract Hermitian matrices, so a real paravector has a Hermitian image. The norm in (A.9) is half the squared Frobenius norm; the determinant in (A.6) may be complex.

The matrix adjugate is

$$\text{adj}(\mathbf{M}) = (\text{tr} \mathbf{M}) \mathbf{I}_2 - \mathbf{M} = \text{mat}(\text{adj}(Z)) = (\mathbf{M}^*)^{\mathbf{H}}. \quad (\text{A.10})$$

It satisfies

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M}) \mathbf{I}_2.$$

Consequently $\mathbf{M}^{-1} = \text{adj}(\mathbf{M}) / \det \mathbf{M}$ when the determinant is nonzero. The determinant product and adjugate order rules are the same in both representations. The quadratic identity (3.31) is precisely Cayley–Hamilton for a 2×2 matrix.

A.4 Matrix amplitudes and column spinors

Let ψ, ψ' be complex paravector amplitudes, with matrix images $\mathbf{M}_\psi, \mathbf{M}_{\psi'}$, and let $Z = Z^{\mathbf{H}}$ be a real observable. For fields these depend on real $(t, \mathbf{r})$. The inner products (8.1) and (8.2) become

$$\begin{aligned} \langle \psi' | \psi \rangle &= \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\mathbf{H}} \mathbf{M}_\psi), \\ \langle \psi' | Z | \psi \rangle &= \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\mathbf{H}} \text{mat}(Z) \mathbf{M}_\psi). \end{aligned}$$

An arbitrary spin basis. Recall the arbitrary constant perpendicular **real unit vectors** $\mathbf{u}, \mathbf{a} \in \mathbb{R}^3$ of (8.5). Set $\Pi_+ = (1 + \mathbf{u} \cdot \boldsymbol{\sigma})/2$ and choose a constant complex paravector ψ' with $\Pi_+ \psi' \neq 0$. Then

$$\begin{Bmatrix} \psi_+ \\ \psi_- \end{Bmatrix} := \frac{1}{\|\Pi_+ \psi'\|} \begin{Bmatrix} \Pi_+ \psi' \\ (\mathbf{a} \cdot \boldsymbol{\sigma}) \Pi_+ \psi' \end{Bmatrix}. \quad (\text{A.11})$$

This constructs (8.5) with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_{\pm} = \pm\psi_{\pm}$ and $\langle \psi_s | \psi_{s'} \rangle = \delta_{ss'}$ for $s, s' = +, -$. The two complex scalar components give

$$\psi = c_+ \psi_+ + c_- \psi_- = (c_+ + c_- \mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+. \quad (\text{A.12})$$

For fields, $c_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$, and (8.3) is $\rho = \|c_+\|^2 + \|c_-\|^2$.

The two-component column. Every column of $\mathbf{M}_{\psi_+}$ has eigenvalue +1 under $\text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma})$, whose +1 eigenspace is one-dimensional. Since $\|\psi_+\| = 1$, its matrix therefore factors as $\mathbf{M}_{\psi_+} = \sqrt{2} \psi_+ \mathbf{v}^H$ for constant columns $\psi_+, \mathbf{v} \in \mathbb{C}^2$ satisfying $\psi_+^H \psi_+ = \mathbf{v}^H \mathbf{v} = 1$. Left multiplication gives $\mathbf{M}_{\psi_-} = \sqrt{2} \psi_- \mathbf{v}^H$, where $\psi_- = \text{mat}(\mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+$. The columns ψ_+, ψ_- are orthonormal. Thus every amplitude in (A.12) has the conventional Pauli column

$$\boldsymbol{\psi} := \frac{1}{\sqrt{2}} \mathbf{M}_{\psi} \mathbf{v} = c_+ \psi_+ + c_- \psi_-. \quad (\text{A.13})$$

Conversely, $\mathbf{M}_{\psi} = \sqrt{2} \boldsymbol{\psi} \mathbf{v}^H$, so this correspondence is one-to-one. For two amplitudes in this fixed basis, substitution cancels the factor of two from the trace pairing:

$$\begin{aligned} \langle \psi' | \psi \rangle &= \boldsymbol{\psi}'^H \boldsymbol{\psi}, \\ \langle \psi' | Z | \psi \rangle &= \boldsymbol{\psi}'^H \text{mat}(Z) \boldsymbol{\psi}. \end{aligned}$$

Left action $T\psi$ maps to $\text{mat}(T)\boldsymbol{\psi}$ for any complex paravector T .

Let $\mathbf{C}$ be the 2×2 matrix with columns ψ_+, ψ_- . Then $\mathbf{C}^H \mathbf{C} = \mathbf{I}_2$ and $\boldsymbol{\psi} = \mathbf{C} \{c_+, c_-\}^T$. For a complex paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, its matrix in this basis is

$$\text{mat}'(Z) := \mathbf{C}^H \text{mat}(Z) \mathbf{C} = \begin{Bmatrix} S + \mathbf{u} \cdot \mathbf{V} & (\mathbf{a} - i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{V} \\ (\mathbf{a} + i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{V} & S - \mathbf{u} \cdot \mathbf{V} \end{Bmatrix}. \quad (\text{A.14})$$

Here the prime denotes only this change of matrix basis. For nonzero norm $\rho = \boldsymbol{\psi}^H \boldsymbol{\psi}$, the real state paravector B and projectors $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ translate to

$$\frac{1}{2} \text{mat}(B) = \boldsymbol{\psi} \boldsymbol{\psi}^H / \rho, \quad (\text{A.15})$$

$$p_{\pm}(\mathbf{u}) = \boldsymbol{\psi}^H \text{mat}(\Pi_{\pm}) \boldsymbol{\psi} / \rho. \quad (\text{A.16})$$

Equations (A.15) and (A.16) are the ordinary density matrix and Born rule.

A.5 Dirac columns

For a Dirac block $\boldsymbol{\Psi} = \{\Psi_1, \Psi_2\}^T$, let $\boldsymbol{\psi}_1, \boldsymbol{\psi}_2 \in \mathbb{C}^2$ be the corresponding columns. Then

$$\mathbf{W}(Z) \boldsymbol{\Psi} \longleftrightarrow \begin{Bmatrix} 0 & \text{mat}(Z) \\ \text{mat}(\text{adj}(Z)) & 0 \end{Bmatrix} \begin{Bmatrix} \boldsymbol{\psi}_1 \\ \boldsymbol{\psi}_2 \end{Bmatrix}.$$

The block norm in (8.30) becomes the ordinary four-component norm:

$$\rho = \boldsymbol{\psi}_1^H \boldsymbol{\psi}_1 + \boldsymbol{\psi}_2^H \boldsymbol{\psi}_2. \quad (\text{A.17})$$

Characteristic polynomial. On one Fourier mode, $s \in \mathbb{C}$ is the spectral parameter and $\square$ its scalar eigenvalue. The two off-diagonal blocks multiply to $-\square \mathbf{I}_2$, so block elimination gives the ordinary matrix determinant

$$\det \begin{Bmatrix} -(s+m)\mathbf{I}_2 & \text{mat}(\text{i}\partial) \\ \text{mat}(\text{i}\partial^*) & -(s+m)\mathbf{I}_2 \end{Bmatrix} = \det((s+m)^2 + \square)\mathbf{I}_2 \quad (\text{A.18})$$

$$= ((s+m)^2 + \square)^2.$$

This verifies (8.40), including multiplicities. The polynomial identity also holds at $s = -m$.

A.6 Four components and the standard Dirac basis

In the arbitrary-axis basis (A.11), write $\Psi_1 = a\psi_+ + b\psi_-$ and $\Psi_2 = c\psi_+ + d\psi_-$, where $a, b, c, d : \mathbb{R}^4 \rightarrow \mathbb{C}$ are the four scalar components. Their columns are $\psi'_1 = \{a, b\}^\top$ and $\psi'_2 = \{c, d\}^\top$; here the prime means $\psi'_k = \mathbf{C}^H \psi_k$. Recall real mass $m > 0$, real charge q , and real potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The mechanical energy and momentum operators are $E = \text{i}\partial_t - qV$ and $\mathbf{p} = -\text{i}\partial_{\mathbf{r}} - q\mathbf{A}$. Their momentum components in this frame are

$$p_1 = \mathbf{a} \cdot \mathbf{p}, \quad p_2 = (\mathbf{u} \times \mathbf{a}) \cdot \mathbf{p}, \quad p_3 = \mathbf{u} \cdot \mathbf{p}.$$

Using (A.14), the coupled paravector equations give

$$(E - p_3)c - (p_1 - \text{i}p_2)d = ma, \quad (\text{A.19})$$

$$-(p_1 + \text{i}p_2)c + (E + p_3)d = mb, \quad (\text{A.20})$$

$$(E + p_3)a + (p_1 - \text{i}p_2)b = mc, \quad (\text{A.21})$$

$$(p_1 + \text{i}p_2)a + (E - p_3)b = md. \quad (\text{A.22})$$

Operators in (A.19)–(A.22) act on the scalar immediately to their right. Their Hamiltonian on $\{a, b, c, d\}^\top$ is

$$H = \begin{Bmatrix} qV - p_3 & -p_1 + \text{i}p_2 & m & 0 \\ -p_1 - \text{i}p_2 & qV + p_3 & 0 & m \\ m & 0 & qV + p_3 & p_1 - \text{i}p_2 \\ 0 & m & p_1 + \text{i}p_2 & qV - p_3 \end{Bmatrix}. \quad (\text{A.23})$$

This is the component image of (8.44).

To identify conventional gamma matrices, put $\sigma_k = \text{mat}(\sigma_k)$ for $k = 1, 2, 3$. These are also the images under mat' of $\mathbf{a} \cdot \boldsymbol{\sigma}$, $(\mathbf{u} \times \mathbf{a}) \cdot \boldsymbol{\sigma}$, and $\mathbf{u} \cdot \boldsymbol{\sigma}$, respectively. Their numerical entries apply to every choice of $\mathbf{u}, \mathbf{a}$. Define

$$\gamma^0 = \begin{Bmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{Bmatrix}, \quad (\text{A.24})$$

$$\gamma^k = \begin{Bmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{Bmatrix}. \quad (\text{A.25})$$

Equations (A.24)–(A.25) obey $(\gamma^0)^2 = \mathbf{I}_4$, $(\gamma^k)^2 = -\mathbf{I}_4$, and $\gamma^j \gamma^k + \gamma^k \gamma^j = 0$ for distinct $j, k = 0, 1, 2, 3$. Thus, in this basis,

$$\mathbf{W}(\text{i}\partial - q\Phi^*) \longleftrightarrow \gamma^0 E - \sum_{k=1}^3 \gamma^k p_k, \quad (\text{A.26})$$

which is the standard chiral Dirac operator with signature $(+, -, -, -)$ [6, p. 83]. Here $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ is the real potential paravector.

The unitary change to the standard Dirac basis commutes with the scalar rest-phase factor explained in (9.1). Together they define the envelope columns

$$\begin{Bmatrix} \phi \\ \chi \end{Bmatrix} = \frac{e^{imt}}{\sqrt{2}} \begin{Bmatrix} \mathbf{I}_2 & \mathbf{I}_2 \\ -\mathbf{I}_2 & \mathbf{I}_2 \end{Bmatrix} \begin{Bmatrix} \psi'_1 \\ \psi'_2 \end{Bmatrix}. \quad (\text{A.27})$$

Here $\phi, \chi \in \mathbb{C}^2$ are the coefficient columns in the same spin basis of the paravector fields $\phi = e^{imt}(\Psi_1 + \Psi_2)/\sqrt{2}$ and $\chi = e^{imt}(\Psi_2 - \Psi_1)/\sqrt{2}$. Before removing the rest phase, the basis change in (A.27) sends (A.23) to

$$H' = \begin{Bmatrix} (qV + m)\mathbf{I}_2 & \sum_k p_k \boldsymbol{\sigma}_k \\ \sum_k p_k \boldsymbol{\sigma}_k & (qV - m)\mathbf{I}_2 \end{Bmatrix}. \quad (\text{A.28})$$

Multiplying the full column by e^{imt} subtracts m from its Hamiltonian: $i\partial_t(e^{imt}\psi) = e^{imt}(i\partial_t - m)\psi$. Thus (A.28) becomes $H' - mI_4$: its diagonal entries are qV and $qV - 2m$. Moving qV into $E = i\partial_t - qV$ gives the exact envelope equations

$$\begin{aligned} E\phi &= \sum_k p_k \boldsymbol{\sigma}_k \chi, \\ (E + 2m)\chi &= \sum_k p_k \boldsymbol{\sigma}_k \phi. \end{aligned}$$

These identify the large and small amplitudes used in the Pauli limit.

For real electric and magnetic components E_k, B_k in the same frame, the squared Dirac field coupling has the explicit image

$$\begin{aligned} \text{mat}'(iqF) &= q \begin{Bmatrix} iE_3 - B_3 & iE_1 + E_2 - B_1 + iB_2 \\ iE_1 - E_2 - B_1 - iB_2 & -iE_3 + B_3 \end{Bmatrix}, \\ \text{mat}'(iqF^*) &= \text{mat}'(iqF)^H. \end{aligned} \quad (\text{A.29})$$

The indexed E_k are field components; unindexed E remains the mechanical energy operator. The two couplings are $-q\mathbf{B} \cdot \boldsymbol{\sigma} \pm iq\mathbf{E} \cdot \boldsymbol{\sigma}$.

A.7 Quaternion subalgebra

Define $q_k = -i\sigma_k$ for $k = 1, 2, 3$, and $\mathbf{q} = \{q_1, q_2, q_3\}$. These are bivectors, not electric charges. Their product follows from the sigma rule:

$$q_j q_k = -\delta_{jk} + \sum_{n=1}^3 \epsilon_{jkn} q_n. \quad (\text{A.30})$$

In particular, $q_k^2 = -1$, $q_1 q_2 = q_3$, and $q_1 q_2 q_3 = -1$. For real a and $\mathbf{A} \in \mathbb{R}^3$, the quaternion

$$Q = a + \mathbf{A} \cdot \mathbf{q} = a - i\mathbf{A} \cdot \boldsymbol{\sigma} \quad (\text{A.31})$$

is a complex paravector with real quaternion coordinates. These elements form the real scalar-bivector subalgebra. For a second such element $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$, expansion gives

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}.$$

Because $q_k^H = -q_k$ and $q_k^* = q_k$, quaternion conjugation is the existing Hermitian conjugation:

$$\begin{aligned} Q^H &= a - \mathbf{A} \cdot \mathbf{q}, \\ Q^* &= Q, \\ \text{adj}(Q) &= Q^H, \\ QQ^H &= Q^H Q = a^2 + \mathbf{A}^2 = \det Q. \end{aligned} \tag{A.32}$$

For nonzero Q , (A.32) gives the inverse $Q^H/(a^2 + \mathbf{A}^2)$. The explicit matrix image is

$$\text{mat}(Q) = \begin{Bmatrix} a - iA_3 & -A_2 - iA_1 \\ A_2 - iA_1 & a + iA_3 \end{Bmatrix}. \tag{A.33}$$

Here $\text{mat}(q_k) = -i \text{mat}(\sigma_k)$. The conjugation and norm identities become

$$\begin{aligned} \text{mat}(Q^H) &= \text{mat}(Q)^H, \\ \text{mat}(Q)^H \text{mat}(Q) &= (a^2 + \mathbf{A}^2) \mathbf{I}_2, \\ \det(\text{mat}(Q)) &= a^2 + \mathbf{A}^2. \end{aligned}$$

A unit quaternion means $a^2 + \mathbf{A}^2 = 1$: it is a **paravector of unit norm** and corresponds to a unitary matrix of determinant one. For **real unit vector** $\mathbf{u}$ and angle θ , let

$$Q = \cos(\theta/2) + \mathbf{u} \cdot \mathbf{q} \sin(\theta/2).$$

Then $Q = R^H$ for $R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$. For a real vector $\mathbf{r}$, the quaternion rotation

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H \tag{A.34}$$

is Rodrigues' formula (4.11) in quaternion form.

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Equation reference

Each displayed formula repeats a numbered result; its right-hand number links to the definition or derivation. Natural units are $\hbar = c = 1$. The linked text supplies domains, component expansions, and approximation conditions. Speculative models are confined to Section 14.

Algebra. Recall complex paravectors $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, with complex scalar S, S' and complex vectors $\mathbf{V}, \mathbf{V}'$. The prime here labels a second element.

$$\begin{aligned} ZZ' &= (SS' + \mathbf{V} \cdot \mathbf{V}') \\ &\quad + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \end{aligned} \tag{3.34}$$

$$Z^H = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} \tag{3.5}$$

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} \tag{3.4}$$

$$\text{adj}(Z) = Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma} \tag{3.6}$$

$$\text{tr } Z = 2 \text{sc}(Z) \tag{3.7}$$

$$\text{re}(Z) = \frac{1}{2}(Z + Z^H) \tag{3.17}$$

$$\text{im}(Z) = (Z - Z^H)/(2i) \tag{3.18}$$

$$\det Z = \text{adj}(Z) Z = S^2 - \mathbf{V}^2 \tag{3.8}$$

$$Z^{-1} = \text{adj}(Z) / \det Z \quad (\det Z \neq 0) \tag{3.12}$$

$$\|Z\|^2 = \text{sc}(Z^H Z). \tag{3.39}$$

The coefficient form is (3.21); product-order and cyclicity rules are (2.3), (2.4), and (3.16). Projectors and the spectral rule are (3.41) and (3.45). Matrix and quaternion images are (A.1) and (A.33).

Rotations and boosts. Here $\mathbf{u}$ is a real unit axis and θ a real angle or rapidity; $\beta = \tanh \theta$ and $\gamma = \cosh \theta$ in the boost formulas.

$$R = \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cos(\theta/2) + i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma} \tag{3.55}$$

$$L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cosh(\theta/2) + \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma} \tag{3.56}$$

$$L = \sqrt{\frac{\gamma+1}{2}} \left(1 + \frac{\gamma\beta}{\gamma+1} \mathbf{u} \cdot \boldsymbol{\sigma} \right) \tag{3.71}$$

$$L^2 = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}) \tag{3.72}$$

$$Z' = R^H Z R \tag{4.10}$$

$$Z' = L^{-1} Z L^{-1}. \tag{4.12}$$

The last line is the passive boost of a real four-vector. Rodrigues' rotation is (4.11); boost components are (4.13)–(4.14). The general determinant-one congruence is (4.8); physical field laws are in Section 11.

Spacetime and particles. Recall real event X , proper time s , coordinate velocity $\mathbf{v}$, and constant mass $m > 0$. The real proper velocity $U = dX/ds$ is a [unit paravector](#) with $UU^{-1} = 1$. The real four-momentum P has mechanical energy E (including rest energy) and spatial momentum $\mathbf{p}$.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \quad (5.1)$$

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 \quad (5.3)$$

$$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2 \quad (6.1)$$

$$U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) \quad (6.4)$$

$$U^*U = 1 \quad (4.5)$$

$$P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma} \quad (6.5)$$

$$E^2 - \mathbf{p}^2 = m^2. \quad (6.6)$$

Here $\gamma = dt/ds = (1 - \mathbf{v}^2)^{-1/2}$. Proper velocity is real, with [determinant one](#); Hermitian unitarity is the separate condition (3.25). For real proper times $s_0 < s_1$ and coordinate times $t_0 < t$,

$$X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds \quad (6.14)$$

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'. \quad (6.15)$$

Clock comparison is (6.16); constant proper acceleration gives (6.17); the future cone is (6.18). Proper acceleration, inertia, and particle action are (6.7), (6.12), and (6.19). Moving-frame velocity and its causal bound are (11.9)–(11.10); clock dilation and measured contraction are (11.11) and (11.12); energy and momentum transform by (11.15)–(11.16).

Electromagnetism. Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. Recall real potentials $V, \mathbf{A}$, real electric and magnetic fields $\mathbf{E}, \mathbf{B}$, the complex field F , real potential paravector Φ , and real charge-current paravector D . The real gauge scalar is S ; q is charge. These fields are functions of real spacetime. Their boosted arguments and components are (11.17) and (11.18)–(11.19).

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad (7.1)$$

$$\partial F = D^* \quad (7.2)$$

$$\square F = \partial^* D^* \quad (7.7)$$

$$\partial \Phi = S + F^* \quad (7.11)$$

$$F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B} \quad (7.19)$$

$$\frac{dP}{ds} = q \operatorname{re}(FU) \quad (7.26)$$

$$T(N) = \frac{1}{2}FN F^H \quad (7.30)$$

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.32)$$

Here N is a real paravector argument of the stress map, u field energy density, $\mathbf{S}$ energy flux, and $\mathbf{J}$ electric current density. Potential components are (7.12)–(7.13); force components are (7.22) and (7.24). The action derivation uses (7.17) and (7.20).

Relativistic waves. Recall real mass $m > 0$, charge q , and potentials $V, \mathbf{A}$. Their real potential paravector is $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. Here $E, \mathbf{p}$ are mechanical differential operators, ψ a complex scalar KG field, and Ψ a two-entry complex paravector Dirac field.

$$E = i\partial_t - qV \quad (8.11)$$

$$\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A} \quad (8.12)$$

$$(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} \quad (8.16)$$

$$[E, \mathbf{p} \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma} \quad (8.17)$$

$$(-E^2 + \mathbf{p}^2 + m^2)\psi = 0 \quad (8.24)$$

$$\mathbf{W}(Z) = \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix} \quad (8.31)$$

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (8.41)$$

Here Z is a complex paravector and $\mathbf{I}$ the block identity. The free equations are (8.20) and (8.34); the two-branch solution is (8.22). Massive Dirac reconstruction is given by (8.39) and (8.56); the squared interacting entries are (8.52) and (8.53).

Low-energy limits. After removing rest energy by (9.1) and (9.12), the complex scalar envelope ψ obeys Schrödinger's equation to leading order:

$$i\partial_t\psi = \left(qV + \frac{\mathbf{p}^2}{2m}\right)\psi. \quad (9.14)$$

The complex paravector envelope ϕ obeys Pauli's equation:

$$i\partial_t\phi = \left(qV + \frac{\mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\right)\phi. \quad (9.9)$$

The recursion and approximation conditions are in Section 9; these are the same leading order for scalar and spinor amplitudes respectively.

Spin and probabilities. Recall a nonzero complex paravector amplitude ψ , real norm density $\rho > 0$, real state paravector B , and [real unit measurement axis](#) $\mathbf{u}$. The real observable Z acts from the left.

$$\langle \psi' | \psi \rangle = \text{sc}(\psi'^H \psi) \quad (8.1)$$

$$\rho = \text{sc}(\psi^H \psi) \quad (8.3)$$

$$B = \psi\psi^H / \rho = 1 + \mathbf{s} \cdot \boldsymbol{\sigma} \quad (10.5)$$

$$p_{\pm}(\mathbf{u}) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}) \quad (10.6)$$

$$\langle Z \rangle_{\psi} = \text{sc}(\psi^H Z \psi) / \rho. \quad (8.4)$$

Here ψ' is a second paravector amplitude and $\mathbf{s}$ the real polarization. The matrix verification is (A.15)–(A.16); spin uncertainty is (10.8).

Magnetic duality and constant-field motion. Recall the real proper velocity $U = dX/ds$, a [unit paravector](#) with $UU^{-1} = 1$. For real magnetic-current paravector M , Maxwell's extension is

$$\partial F = D^* + iM^*. \quad (13.4)$$

For real electric and magnetic charges q, q' , the force is

$$\frac{dP}{ds} = \text{re}((q - iq')FU). \quad (13.1)$$

For constant F and real future initial momentum with mass $m > 0$, define the complex [unit paravector](#)

$$T(s) = \exp\left(\frac{(q - iq')s}{2m}F\right). \quad (13.11)$$

The real future mass-shell solution is

$$P(s) = T(s)P(0)T(s)^H. \quad (13.12)$$

This classical formalism retains real spacetime. The separate complex-coordinate hypotheses and their failed ordinary limits are tested in [Section 14](#).